場の理論　　Apr 8

量子力学 ⊕ 特殊相対論

1. 量子力学の復習

Lagrangian　$L = L(q, \dot{q})$

$$p_k = \frac{\partial L}{\partial \dot{q}_k} = m\dot{q}_k$$

Hamiltonian　$H(p_k, q_k) = \sum_k p_k \dot{q}_k - L(q_k, \dot{q}_k)$　(Legendre 変換)

$$= \sum_k \frac{1}{2m} p_k^2 - V(q_k)$$

Poisson bracket　$\{F, H\}_{PB} = \sum_k \left\{ \frac{\partial F}{\partial q_k}\frac{\partial H}{\partial p_k} - \frac{\partial H}{\partial q_k}\frac{\partial F}{\partial p_k} \right\} = -\{H, F\}_{PB}$

$$\{q_k, p_l\}_{PB} = \delta_{kl} \qquad \{p_k, p_l\}_{PB} = \{q_k, q_l\}_{PB} = 0$$

$$\dot{p}_k = \{p_k, H\}_{PB} = -\frac{\partial V}{\partial q_k}$$

$$\dot{q}_k = \{q_k, H\}_{PB} = \frac{1}{m} p_k$$

量子化

$$\{F, H\}_{PB} \Rightarrow \frac{1}{i\hbar}[\hat{F}, \hat{H}] = \frac{1}{i\hbar}(\hat{F}\hat{H} - \hat{H}\hat{F})$$

$$\{q_k, p_k\}_{PB} = \delta_{kl} \Rightarrow \begin{cases} [\hat{q}_k, \hat{p}_l] = i\hbar\delta_{kl} \\ [\hat{q}_k, \hat{q}_l] = 0 \\ [\hat{p}_k, \hat{p}_l] = 0 \end{cases}$$

$$\hat{\dot{p}}_k = \frac{1}{i\hbar}[\hat{p}_k, \hat{H}] = -\frac{\partial}{\partial q_k} V(q_k)$$

$$\hat{\dot{q}}_k = \frac{1}{i\hbar}[\hat{q}_k, \hat{H}] = \frac{1}{m}\hat{p}_k$$

$\hat{p}_k, \hat{q}_k \Rightarrow$ 線形演算子 (linear operator)

$|\psi\rangle$ 状態ベクトル

$\frac{d}{dt}|\psi\rangle$ Heisenberg picture

Schrödinger picture $\frac{d}{dt}\hat{p}_s = \frac{d}{dt}\hat{q}_s = 0$ $\quad i\hbar\frac{\partial}{\partial t}|\psi_s\rangle = \hat{H}|\psi_s\rangle$

一般に $\langle\psi_H, \hat{O}(t)\psi_H\rangle \equiv \langle\psi_s(t), \hat{O}(0)\psi_s(t)\rangle$

$$i\hbar\dot{\hat{O}}(t) = [\hat{O}(t), \hat{H}] \qquad \hat{H}^+ = \hat{H}$$

あるいは

$$\hat{O}(t) = e^{i\hat{H}t/\hbar}\,\hat{O}(0)\,e^{-i\hat{H}t/\hbar}$$

表示 (representation)

$\hat{q}_s$ を対角化する表示 $\quad \langle q_s|\hat{q}_s = q_s\langle q_s|$

$$\langle q_s|\hat{p}_s = \frac{\hbar}{i}\frac{\partial}{\partial q_s}\langle q_s|$$

$\langle q_s|\psi_s(t)\rangle \equiv \psi(t, q_s)$ 波動関数

$$i\hbar\frac{\partial}{\partial t}\psi(t,q_s) = \langle q_s|i\hbar\frac{\partial}{\partial t}|\psi_s(t)\rangle = \langle q_s|\hat{H}(\hat{p}_s,\hat{q}_s)|\psi_s(t)\rangle = H\left(\frac{\hbar}{i}\frac{\partial}{\partial q_s}, q_s\right)\langle q_s,\psi_s(t)\rangle = H\left(\frac{\hbar}{i}\frac{\partial}{\partial q_s}, q_s\right)\psi(t,q_s)$$

$\psi(t,q_s)$ = Schrödinger 波動関数

= 無限次元のベクトル

エネルギーの固有ベクトル

$$H\left(\frac{\hbar}{i}\frac{\partial}{\partial q_s}, q_s\right)\varphi_n(q_s) = E_n\varphi_n(q_s)$$

$$\int\varphi_m(q_s)^*\varphi_n(q_s)\,d^3q = \delta_{mn}$$

$$\psi(t,q_s) = \sum_{n=1}^{\infty} c_n(0)\,e^{-iE_nt/\hbar}\,\varphi_n(q_s) = \sum_{n=1}^{\infty} c_n(t)\,\varphi_n(q_s)$$

$$\psi(t,q_s) \longleftrightarrow \begin{pmatrix} c_1(t) \\ c_2(t) \\ \vdots \end{pmatrix}$$

$\rho(q_s) = \psi(t,q_s)\psi^*(t,q_s)$: 粒子が q_s に存在する確率密度 (M. Born)

$$\int d^3q\,\rho(q_s) = \sum_{n=1}^{\infty}|c_n(t)|^2 < \infty$$

(Hilbert space)

一般に任意の完全系

$\{\phi_n(q_s)\}$ $\quad \int\phi_n(q_s)\phi_m(q_s)\,d^3q = \delta_{mn}$

$$\psi(t,q_s) = \sum_{n=1}^{\infty} a_n(t)\,\phi_n(q_s) = \sum_{n=1}^{\infty} c_n(t)\,\varphi_n(q_s)$$

$$c_n(t) = \int d^3q\;\varphi_n(q_s)^*\psi(t,q_s) = \sum_m\int d^3q\;\varphi_n(q_s)^*\phi_m(q_s)\,a_m(t) \equiv \sum_m\langle\varphi_n|\phi_m\rangle\,a_m(t)$$

$$\begin{pmatrix} a_1 \\ a_2 \\ \vdots \end{pmatrix} \longrightarrow \begin{pmatrix} c_1 \\ c_2 \\ \vdots \end{pmatrix}$$ への unitary 変換

この記法を使うと

$\psi_n(q_0) = \langle q_0 | E_n \rangle$

ユニタリー行列 $\longleftrightarrow$ $\{\psi_n(q_0)\}$ の正規直交性 (Dirac)

Schrödinger 方程式

$i\hbar \frac{\partial}{\partial t}\psi(t, q_0) = \hat{H}\psi(t, q_0)$

初期条件 $\psi(0, q_0) = \phi_n(q_0)$

形式解 $\psi(t, q_0) = e^{-i\hat{H}t/\hbar}\phi_n(q_0)$

$\int \phi_m(q_0)^* \psi(t, q_0)\, d^3q$ = 時間 t に $\phi_m(q_0)$ にある確率振幅

$= \int d^3q\, \phi_m(q_0)^* e^{-i\hat{H}t/\hbar}\phi_n(q_0) = \langle m | e^{-i\hat{H}t/\hbar} | n \rangle$ $\quad UU^\dagger = U^\dagger U = 1$

$= \int d^3q'\, d^3q\, \langle m | q' \rangle \langle q' | e^{-i\hat{H}t/\hbar} | q \rangle \langle q | n \rangle$ $\quad \sum_n \langle q' | E_n \rangle \langle E_n | q \rangle = \delta(q'-q)$

時間発展の演算子 (evolution operator)

$\langle q_0' | e^{-i\hat{H}t/\hbar} | q_0 \rangle = e^{-iH(\frac{\hbar}{i}\frac{\partial}{\partial q_0'}, q_0')t/\hbar} \langle q_0' | q_0 \rangle = e^{-iH(\frac{\hbar}{i}\frac{\partial}{\partial q_0'}, q_0')t/\hbar}\, \delta^3(q_0' - q_0)$

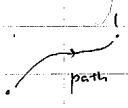
$= \sum_n e^{-iH(\frac{\hbar}{i}\frac{\partial}{\partial q_0'}, q_0')t/\hbar}\, \phi_n(q_0')\phi_n^*(q_0)$

$= \sum_{\text{all path}} \exp\left[\frac{i}{\hbar}\int_0^t L(q_0, \dot{q}_0)\, dt\right]$

$\int L\, dt = \int (p\dot{q} - H)\, dt = \oint p\, dq - E \cdot t$

$= \sum_{\text{all path}} \exp\left[\frac{i}{\hbar} S(q_0', t; q_0, 0)\right]$ Feynman の経路積分 (path integral)

$S(q_0', t; q_0, 0) = \int_0^t L(q_0, \dot{q}_0)\, dt$ = 古典的作用 (classical action)



Schrödinger 方程式を解く = path について足す

摂動展開 (Perturbation)

遷移振幅 $\langle m | e^{-i\hat{H}t/\hbar} | n \rangle$

$\hat{H} = \hat{H}_0 + \hat{H}_I$

自由粒子　相互作用 (interaction)

今、$f(t) \equiv e^{i\hat{H}_0 t/\hbar} e^{-i\hat{H}(t-t_i)/\hbar} e^{-i\hat{H}_0 t_i/\hbar}$ を考える

$f(t_i) = 0$

$f'(t) = e^{i\hat{H}_0 t/\hbar}\left[\frac{i\hat{H}_0}{\hbar} - \frac{i\hat{H}}{\hbar}\right] e^{-i\hat{H}(t-t_i)/\hbar} e^{-i\hat{H}_0 t/\hbar} = e^{i\hat{H}_0 t/\hbar}\left(-\frac{i}{\hbar}\hat{H}_I\right) e^{-i\hat{H}_0 t/\hbar} f(t) = -\frac{i}{\hbar}\hat{H}_I(t) f(t)$ $\quad (*)$

$\hat{H}_I(t) = e^{i\hat{H}_0 t/\hbar}\hat{H}_I(\hat{q}_0, \hat{p}_0) e^{-i\hat{H}_0 t/\hbar}$

$(*)$ の形式解

$f(t) = 1 - \frac{i}{\hbar}\int_{t_i}^t \hat{H}_I(t) f(t)\, dt$ から $f(t) = T \exp\left[-\frac{i}{\hbar}\int_{t_i}^t \hat{H}_I(t)\, dt\right] f(t_i)$

↑ 時間順序積 (time ordering)

∵) $f(t) = 1 - \frac{i}{\hbar}\int_{t_i}^{t} \hat{H}_I(t)\, f(t)\, dt = 1 - \frac{i}{\hbar}\int_{t_i}^{t} \hat{H}_I(t)\, dt + \left(\frac{-i}{\hbar}\right)^2 \int_{t_i}^{t} dt' \hat{H}_I(t') \int_{t_i}^{t'} dt'' \hat{H}_I(t'')\, dt'' + \cdots$

$= 1 - \frac{i}{\hbar}\int_{t_i}^{t} \hat{H}_I(t)\, dt + \frac{1}{2!}\left(\frac{-i}{\hbar}\right)^2 T\left[\int_{t_i}^{t} dt' \hat{H}_I(t') \int_{t_i}^{t} \hat{H}_I(t'')\, dt'' + \cdots\right]$

$= T\exp\left[-\frac{i}{\hbar}\int_{t_i}^{t} \hat{H}_I(t')\, dt'\right]$

※ iteration の形で解ける

但 $T\,\hat{H}_I(t')\hat{H}_I(t'') \equiv \hat{H}_I(t')\hat{H}_I(t'')$ for $t' > t''$

$\equiv \hat{H}_I(t'')\hat{H}_I(t')$ for $t'' > t'$

一般に U-matrix と呼ばれるのは

$$U(t_f, t_i) = f(t_f) = T\exp\left[-\frac{i}{\hbar}\int_{t_i}^{t_f} \hat{H}_I(t)\, dt\right]$$

(t_f: final)

S 行列 (scattering matrix) $\quad S = \lim_{\substack{t_f\to\infty \\ t_i\to-\infty}} T\exp\left[-\frac{i}{\hbar}\int_{t_i}^{t_f} \hat{H}_I(t)\, dt\right]$

一般に $|i\rangle$ から $|f\rangle$ への遷移行列要素は

$\langle f|S|i\rangle$

但

$$\langle f|e^{-i\hat{H}(t_f - t_i)/\hbar}|i\rangle = e^{-\frac{i}{\hbar}E_f t_f}\cdot\langle f|S|i\rangle e^{\frac{i}{\hbar}E_i t_i} \quad \left(\begin{matrix} t_f\to\infty \\ t_i\to-\infty \end{matrix}\right.$$

為

$$\hat{H}_I(t) = e^{i\hat{H}_0 t/\hbar}\hat{H}_I(\hat{q}(0))e^{-i\hat{H}_0 t/\hbar} = \hat{H}_I(\hat{q}(t)) \quad \to \quad \hat{q}(t) = e^{i\hat{H}_0 t/\hbar}\hat{q}(0)e^{-i\hat{H}_0 t/\hbar}$$

$$\dot{\hat{q}}(t) = \frac{i}{\hbar}[\hat{H}_0, \hat{q}(t)]$$

全ての演算子は自由粒子の時間発展をする (相互作用表示)

例. 最低次の摂動

$$\langle f|S|i\rangle = \int dt \left(\frac{-i}{\hbar}\right)\langle f|e^{i\hat{H}_0 t/\hbar}H_I(\hat{q}(0))e^{-i\hat{H}_0 t/\hbar}|i\rangle$$

$$= \left(\frac{-i}{\hbar}\right)\int_{t_i}^{t_f} dt\, e^{i(E_f - E_i)t/\hbar}\langle f|H_I(\hat{q}(0))|i\rangle$$

$$= \left(\frac{-i}{\hbar}\right)2\pi\,\delta\left(\frac{E_f - E_i}{\hbar}\right)\int d^3q\, \varphi_f(q)^* H_I(q)\varphi_i(q)$$

$$= (-2\pi i)\,\delta(E_f - E_i)\int d^3q \cdots$$

$\delta(bx) = \frac{1}{b}\delta(x)$

$\delta(f(x)) = \frac{1}{|f'(x_i)|}\delta(x - x_i)$

$$\frac{|\langle f|S|i\rangle|^2}{T} = \left(\frac{2\pi}{\hbar}\right)\delta(E_f - E_i)\left|\int d^3q\, \varphi_f(q)\, H_I(q)\, \varphi_i(q)\right|^2 \quad \text{Fermi's golden rule}$$

$t_f - t_i$: 全時間

単位時間当りの遷移確率

但 $\int_{t_i}^{t_f} e^{i(E_f - E_i)t/\hbar} = t_f - t_i$ for $E_f = E_i$

$= 2\pi\,\delta\left(\frac{0}{\hbar}\right)$

Hartree-Fock 近似

May 13

$$\hat{\mathcal{H}} = \int d^3x\, \hat{\psi}^+(x) H(x) \hat{\psi}(x) + \frac{1}{2}\iint \hat{\psi}^+(x_1)\hat{\psi}^+(x_2) U(x_1,x_2) \hat{\psi}(x_1)\hat{\psi}(x_2) d^3x_1 d^3x_2$$

$H(x) = \frac{p^2}{2m} + V(x)$ ↑ 相互作用

$\hat{\psi}(x) = \sum_n \hat{a}_n \varphi_n(x)$ 状態 $|n_1, n_2 \cdots\rangle$

$\{\varphi_n\}$ の choice に関して

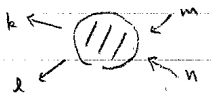
$\hat{\mathcal{H}} = \hat{\mathcal{H}}_0 + \hat{\mathcal{H}}_I$ $\langle \cdots n_2 n_1 | \hat{\mathcal{H}} | n_1, n_2 \cdots \rangle$ これを最小にする

$\hat{\mathcal{H}}_0 = \sum_{k,l} h_{kl} \hat{a}_k^+ \hat{a}_l$ $\langle \cdots n_2 n_1 | \hat{\mathcal{H}}_0 | n_1, n_2 \cdots \rangle = \sum_k h_{kk} n_k$

$h_{kk} = \varphi_k^* \left[-\frac{\hbar^2}{2m}\vec{\nabla}^2 + V(\vec{x})\right] \varphi_k$

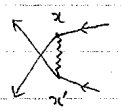
$\hat{n}_k = \hat{a}_k^+ \hat{a}_k$ の固有値

$\hat{\mathcal{H}}_I = \frac{1}{2}\sum_{klmn} u_{klmn} \hat{a}_k^+ \hat{a}_l^+ \hat{a}_m \hat{a}_n$





fermi 統計



$$\langle \cdots n_2 n_1 | \hat{\mathcal{H}}_I | n_1, n_2 \cdots \rangle = \frac{e^2}{2}\int d^3x\, d^3x' \left[\frac{\rho(x)\rho(x')}{|\vec{x}-\vec{x}'|} - \frac{|\rho(x,x')|^2}{|\vec{x}-\vec{x}'|}\right]$$

$$\rho(x,x') = \langle \cdots n_2 n_1 | \hat{\psi}^+(x)\hat{\psi}(x') | n_1, n_2 \cdots \rangle = \sum_l n_l \varphi_l^*(x)\varphi_l(x') = \rho(x',x)^* \qquad \rho(x) = \rho(x,x)$$

変分法

$\varphi_l(x)$ を適当に選んで $\langle \cdots n_2 n_1 | \hat{\mathcal{H}} | n_1, n_2 \cdots \rangle$ を最小にする

但し

$$\int \varphi_k(x)^* \varphi_l(x)\, d^3x = \delta_{kl}$$ を課す

Lagrange の未定定数法

$$\delta\left[\langle \cdots n_2 n_1 | \hat{\mathcal{H}} | n_1, n_2 \cdots \rangle - \lambda_{kl}\left(\int \varphi_k^*(x)\varphi_l(x) d^3x - \delta_{kl}\right)\right] = 0$$

↑ φ_k^* に関するもの

$$\Rightarrow \left[-\frac{\hbar^2}{2m}\vec{\nabla}^2 + V(x)\right]\varphi_l(\vec{x}) + e^2\left[\int d^3x' \frac{\rho(x')}{|\vec{x}-\vec{x}'|}\varphi_l(\vec{x}) - \int d^3x' \frac{\rho(x,x')}{|\vec{x}-\vec{x}'|}\varphi_l(x')\right] = \varepsilon_l \varphi_l(\vec{x})$$

但し $\lambda_{kl} = \lambda_{lk}^*$ を対角化するように φ_l を Unitary 変換して $\lambda_{kl} = \varepsilon_l \delta_{kl}$ とした

このとき $\rho(x'x) = \sum_j \varphi_j^*(x')\varphi_j(x)$ は不変

$\Rightarrow \varphi_l(x)$ に関して self-consistent に解く.

$\Rightarrow \hat{\Psi}(x) = \sum_k \hat{a}_k \varphi_k(x)$ を用いて

波動関数 $\Psi(\vec{x}_1 \cdots \vec{x}_n) = \frac{1}{\sqrt{n!}} \langle 0 | \hat{\Psi}(x_1) \cdots \hat{\Psi}(x_n) | n_1, n_2 \cdots \rangle$ と固有値 ε_k が求まる　　$\sum n_i = n$

但 $E = \sum_{k=1}^{n} \varepsilon_k$

経路積分 (Pass integral)

Evolution operator (Schrödinger 表示)

$$\langle q_f | e^{-i\hat{H}(\hat{p},\hat{q})t/\hbar} | q_i \rangle$$

時間を N コの区間 (mesh) に切ると $\Delta t = t/N$ として

$$= \langle q_f | e^{-i\hat{H}\Delta t/\hbar} e^{-i\hat{H}\Delta t/\hbar} \cdots e^{-i\hat{H}\Delta t/\hbar} | q_i \rangle \quad (\text{final} \to \text{initial})$$

$$= \left(\prod_{k=1}^{N-1} \int d^3 q_k \right) \langle q_f | e^{-i\hat{H}\Delta t/\hbar} | q_{N-1} \rangle \langle q_{N-1} | e^{-i\hat{H}\Delta t/\hbar} | q_{N-2} \rangle \langle q_{N-2} | \cdots \langle q_1 | e^{-i\hat{H}\Delta t/\hbar} | q_i \rangle$$

次に $\langle q_1 | e^{-i\hat{H}(\hat{p},\hat{q})\Delta t/\hbar} | q_i \rangle \fallingdotseq \langle q_1 | 1 - i\hat{H}(\hat{p},\hat{q})\Delta t/\hbar | q_i \rangle$

$$= \int \frac{d^3 p_1}{(2\pi\hbar)^3} \langle q_1 | p_1 \rangle \langle p_1 | 1 - i\hat{H}(\hat{p},\hat{q})\Delta t/\hbar | q_i \rangle$$

$$= \int \frac{d^3 p_1}{(2\pi\hbar)^3} \langle q_1 | p_1 \rangle \left[1 - iH(p_1, q_i)\Delta t/\hbar \right] \langle p_1 | q_i \rangle \quad (H: \text{c数})$$

$$\fallingdotseq \int \frac{d^3 p_1}{(2\pi\hbar)^3} \langle q_1 | p_1 \rangle \langle p_1 | q_i \rangle e^{-iH(p_1,q_i)\Delta t/\hbar}$$

$$= \int \frac{d^3 p_1}{(2\pi\hbar)^3} e^{\left\{ i\vec{p}_1(\vec{q}_1 - \vec{q}_i)/\hbar - iH(p_1,q_i)\Delta t/\hbar \right\}} \qquad (\because) \langle q_1 | p_1 \rangle = e^{i\vec{p}_1\vec{q}_1/\hbar}$$

すなわち

$$\langle q_f | e^{-i\hat{H}(\hat{p},\hat{q})t/\hbar} | q_i \rangle \fallingdotseq \left(\prod_{k=1}^{N-1} \int \frac{d^3 p_k d^3 q_k}{(2\pi\hbar)^3} \right) \int \frac{d^3 p_N}{(2\pi\hbar)^3} \exp \left\{ \frac{i}{\hbar} \left[\vec{p}_N (\vec{q}_1 - \vec{q}_{N-1})/\Delta t - H(p_N, q_f) \right.\right.$$
$$+ \vec{p}_{N-1} (\vec{q}_{N-1} - \vec{q}_{N-2})/\Delta t - H(p_{N-1}, q_{N-1})$$
$$+ \cdots$$
$$\left.\left. + \vec{p}_1 (\vec{q}_1 - \vec{q}_i)/\Delta t - H(p_1, q_i) \right] \Delta t \right\}$$

$q_{N-1} = q(t_{N-1})$ と読みかえて

$$\underset{N\to\infty}{\Rightarrow} \int Dq\,Dp\,\exp\left\{\frac{i}{\hbar}\int_{t_i}^{t_f}\underbrace{\left[\vec{p}(t)\,\dot{\vec{q}}(t) - H(p(t),q(t))\right]}_{L(q,\dot{q})}dt\right\}$$

但 $\begin{cases} q(t_f) = q_f \\ q(t_i) = q_i \end{cases}$ (R. Feynman)

ここで $Dq\,Dq = \dfrac{\Pi}{t}\dfrac{d^3q(t)\,d^3p(t)}{(2\pi\hbar)^3}$ $\qquad \int_{t_i}^{t_f}\left[\vec{p}(t)\cdot\dot{\vec{q}}(t) - H(p(t),q(t))\right]dt$ = 古典的作用

↗ 時間を mesh に切って積

↑ Liouville 測度

Evolution operator = 古典的作用/$\hbar$ を位相因子として、Liouville 測度で積分したもの。

次に $H(p,q) = \dfrac{1}{2m}\vec{p}^{\,2} + V(\vec{q})$ の場合

$$\int \frac{d^3p_l}{(2\pi\hbar)^3}\exp\left\{\frac{i}{\hbar}\left[\vec{p}_l\cdot(\vec{q}_{l+1}-\vec{q}_l) - \frac{1}{2m}p_l^2\,\Delta t\right]\right\}$$

$$= \int \frac{d^3p_l}{(2\pi\hbar)^3}\exp\left\{-\left(\frac{i}{\hbar}\right)\frac{\Delta t}{2m}\left(\vec{p}_l - \frac{m}{\Delta t}(\vec{q}_{l+1}-\vec{q}_l)\right)^2\right\}\exp\left\{\left(\frac{i}{\hbar}\right)\left(\frac{m}{2}\right)\left(\frac{\vec{q}_{l+1}-\vec{q}_l}{\Delta t}\right)^2\Delta t\right\}$$

$$= \left(\frac{m}{2\pi i\hbar\,\Delta t}\right)^{3/2}\exp\left\{\left(\frac{i}{\hbar}\right)\left(\frac{m}{2}\right)\left(\frac{\vec{q}_{l+1}-\vec{q}_l}{\Delta t}\right)^2\Delta t\right\}$$

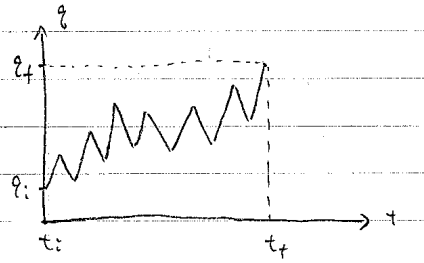
$$\langle q_f|e^{-i\hat{H}t/\hbar}|q_i\rangle = \left(\frac{m}{2\pi i\hbar\Delta t}\right)^{3/2}\int\prod_{l=1}^{N-1}\left(\frac{m}{2\pi i\hbar\Delta t}\right)^{3/2}d^3q_l\,\exp\left\{\left(\frac{i}{\hbar}\right)\left[\frac{m}{2}\left(\frac{\vec{q}_f-\vec{q}_{N-1}}{\Delta t}\right)^2 - V(q_{N-1}) + \cdots + \frac{m}{2}\left(\frac{q_1-q_i}{\Delta t}\right)^2 - V(q_1)\right]\Delta t\right\}$$

$$\underset{N\to\infty}{\Rightarrow} \int \mathcal{D}q\,\exp\left\{\left(\frac{i}{\hbar}\right)\int_{t_i}^{t_f}\left[\frac{m}{2}\dot{\vec{q}}(t)^2 - V(q(t))\right]\right\}$$

$$= \int \mathcal{D}q\,\exp\left[\frac{i}{\hbar}\int_{t_i}^{t_f}L(q(t),\dot{q}(t))\,dt\right]$$

↑ Lagrangian

図式化すると



q_i から q_f への全ての道に関する足し上げ

$$\sum_{\text{path}} e^{\frac{i}{\hbar}S(q_f\,q_i)/\hbar}$$

例) 自由粒子

$$\langle q_f | e^{-i\frac{1}{2m}\vec{P}^2 t/\hbar} | q_i \rangle = \int \frac{d^3p}{(2\pi\hbar)^3} \langle \vec{q}_f | \vec{p} \rangle \langle \vec{p} | e^{-\frac{i}{2m}\hat{\vec{p}}^2 t/\hbar} | \vec{q}_i \rangle$$

$$= \int \frac{d^3p}{(2\pi\hbar)^3} e^{-\frac{i}{2m}\vec{p}^2/\hbar} \underbrace{\langle q_f | p \rangle \langle p | q_i \rangle} e^{i\vec{p}(\vec{q}_f - \vec{q}_i)/\hbar}$$

$$= \left(\frac{m}{2\pi i\hbar t}\right)^{3/2} e^{\frac{i}{\hbar}\frac{m}{2}\frac{(q_f - q_i)^2}{t}}$$

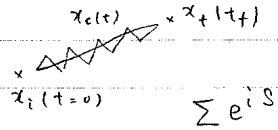
すなわち mesh に切らなくても 正値に定まる

Report

自由粒子の例を無限個の mesh に切って経路積分の結果として求めよ

$$\langle x_f | e^{-i\hat{H}t/\hbar} | x_i \rangle = \sum_{\text{path}} e^{iS(x_f, x_i; t)/\hbar}$$

▷ May 20



WKB 半古典近似

$$\sum_{\delta x} e^{iS(x_c(t)+\delta x)/\hbar}$$

↑ ゆらぎ

$$S(x_c + \delta x) = S(x_c) + \underbrace{S'(x_c)\delta x}_{0} + \frac{1}{2}S''(x_c)\delta x \delta x + \cdots$$

$$= J(x_c) e^{iS(x_c)/\hbar}$$

$$\sum_{\psi\psi^*} e^{iS(\psi\psi^*; t)/\hbar}$$

特殊相対論

$x^\mu = (ct, \vec{x})$

時空間の線素 $ds^2 = (dx^0)^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 = g_{\mu\nu} dx^\mu dx^\nu$

$\mu, \nu = 0, 1, 2, 3$

$k, l = 1, 2, 3$

$g_{\mu\nu} = (1, -1, -1, -1) = g^{\mu\nu}$ (Minkovski 空間)

$g_{\mu\nu} g^{\nu\lambda} = \delta_\mu{}^\lambda$ $\quad dx_\mu = g_{\mu\nu} dx^\nu$ $\quad dx^\nu$ contravariant vector

dx_μ covariant vector

Lorentz 変換 = 座標の (実) 同次線形変換

$x'^\mu = a^\mu{}_\nu x^\nu$ であって ds^2 を変えない

↑ 定数

$ds^2 = g_{\mu\nu}\,dx^\mu dx^\nu = g_{\mu\nu}\,dx'^\mu dx'^\nu = g_{\mu\nu}\,a^\mu{}_\lambda\, a^\nu{}_\rho\, dx^\lambda dx^\rho$

$g_{\lambda\rho} = g_{\mu\nu}\,a^\mu{}_\lambda\, a^\nu{}_\rho$

$G = (g_{\lambda\rho})$

$A = (a^\mu{}_\lambda)$ $\quad G' = A^T G A$ $\quad \det G' = \det A^T \det G \det A \Rightarrow (\det A)^2 = 1$

$\det A = -1$

$A = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$ 時間反転 $\begin{cases} t \to -t \\ \vec{x} \to \vec{x} \end{cases}$

$A = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$ 空間反転 parity $\begin{cases} t \to t \\ \vec{x} \to -\vec{x} \end{cases}$

$\det A = 1$ proper Lorentz 変換

(無限小変換の重ね合わせで生成される

$\det A = -1$ Inproper Lorentz 変換

($A = 1$ から出発して無限小変換の重ね合わせでは生成されない

$\det A = 1$

$(dx^0)^2 - (d\vec{x})^2 = (dx^{0\prime})^2 - (d\vec{x}')^2$

↓ 回転

$= (dx^{0\prime})^2 - (dx^{1\prime})^2 - (dx^{2\prime})^2 - (dx^{3\prime})^2$

$(dx^0)^2 - (dx^1)^2 = (dx^{0\prime})^2 - (dx^{1\prime})^2$

$(i\,dx^0)^2 + (dx^1)^2 = (i\,dx^{0\prime})^2 + (dx^{1\prime})^2$

Lorentz 変換は一般に

$dx^{0\prime} = a^0{}_0\,dx^0 + a^0{}_1\,dx^1$

$dx^{1\prime} = a^1{}_0\,dx^0 + a^1{}_1\,dx^1$

$\begin{cases} i\,dx^{0\prime} = \cos\bar{\theta}\,(i\,dx^0) + \sin\bar{\theta}\,dx^1 \\ i\,dx^{1\prime} = -\sin\bar{\theta}\,(i\,dx^0) + \cos\bar{\theta}\,dx^1 \end{cases}$

ここで $\bar{\theta} = -i\theta$ とおくと

$dx^{0\prime} = \cosh\theta\,dx^0 - \sinh\theta\,dx^1$

$dx^{1\prime} = \sinh\theta\,dx^0 + \cosh\theta\,dx^1$

④ 回転が一般の Lorentz 変換

無限小変換は $|\theta| \ll 1$

$dt' = \left(1 + \frac{\theta^2}{2}\right)dt - \frac{\theta}{c}\,dx'$

$dx'' = -c\theta\,dt - \left(1 + \frac{\theta^2}{2}\right)dx'$

cf $\begin{cases} dx' = dx' - v\,dt \\ dt' = dt \end{cases}$ Galilei 変換

$\cosh\theta = \frac{1}{\sqrt{1-\beta^2}}$

$\Rightarrow \beta = \frac{v}{c} \ll 1$ では $\theta \approx \beta = \frac{v}{c}$

$\sinh\theta = \frac{\beta}{\sqrt{1-\beta^2}}$ とおくと $\beta = \frac{v}{c}$ となる

一般に 2つの vector X^μ Y^μ の内積は不変 $X^\mu Y_\mu = X'^\mu Y'_\mu$

Maxwell 方程式

$\vec{E}$ $\vec{B}$ $A_\mu(x)$ gauge potential . gauge 場

$g_{\mu\nu} dx^\mu dx^\nu = dx^\mu dx_\nu$ invariant

作用 (action)

$$\mathcal{L} = -\frac{1}{4}(\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial^\mu A^\nu - \partial^\nu A^\mu)$$

$F_{\mu\nu}$ $F^{\mu\nu}$

$A^\mu = g^{\mu\nu} A_\nu$

$A^0 = A_0$, $A^k = -A_k$

変換性は

$x'^\mu = a^\mu{}_\nu x^\nu$

$\partial'_\mu = g_{\mu\nu}\partial'^\nu = g_{\mu\nu} a^\nu{}_\lambda \partial^\lambda = \boxed{g_{\mu\nu} a^\nu{}_\lambda g^{\lambda\alpha}} \partial_\alpha$ ($\tilde{a}_\mu{}^\alpha$)

$dx'^\mu = a^\mu{}_\lambda dx^\lambda$

$g_{\mu\nu} dx'^\mu dx'^\nu = dx^\mu dx_\nu$

$dx'_\mu = g_{\mu\nu} a^\nu{}_\lambda dx^\lambda$

$A'^\mu(x') = a^\mu{}_\nu A^\nu(x)$ (定義)

$\phi'(x') = \phi(x)$

更に

$d^4x' = |\det A| d^4x = d^4x$

テンソル $A_{\mu\nu} B^{\mu\nu} = g_{\mu\lambda} g_{\nu\rho} A^{\lambda\rho} A^{\mu\nu}$ 変換則 $A'^{\lambda\rho} = a^\lambda{}_\alpha a^\rho{}_\beta A^{\alpha\beta}$

$A_{\mu_1\mu_2\cdots} B^{\mu_1\mu_2\cdots} = g_{\mu_1\nu_1} g_{\mu_2\nu_2} \cdots A^{\nu_1\nu_2\cdots} B^{\mu_1\mu_2\cdots}$

$A'^{\nu_1\nu_2\cdots} = a^{\nu_1}{}_{\lambda_1} a^{\nu_2}{}_{\lambda_2} \cdots A^{\lambda_1\lambda_2\cdots}$

$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ field strength tensor $F_{\mu\nu} = -F_{\nu\mu}$

作用 $S = \int \mathcal{L} d^4x = -\frac{1}{4}\int F_{\mu\nu} F^{\mu\nu} d^4x$ Lorentz 不変である

運動方程式

$$\delta S = -\frac{1}{4}\delta \int (\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial^\mu A^\nu - \partial^\nu A^\mu) d^4x = 0$$

$$= -\frac{1}{2}\int \delta(\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial^\mu A^\nu - \partial^\nu A^\mu) d^4x = 0$$

$$= -\int \partial_\mu \delta A_\nu (\partial^\mu A^\nu - \partial^\nu A^\mu) d^4x$$

$$= \int \delta A_\nu \underbrace{\partial_\mu (\partial^\mu A^\nu - \partial^\nu A^\mu)}_{=0} d^4x \quad \rightarrow \partial_\mu(\partial^\mu A^\nu - \partial^\nu A^\mu) = 0$$

$\partial_\mu(\partial^\mu A^\nu - \partial^\nu A^\mu) = \partial_\mu\partial^\mu A^\nu - \partial_\mu\partial^\nu A^\mu = 0 \quad \leftarrow eJ^\nu$

$\nu=0$ 成分 $\quad \vec{\nabla}\cdot\vec{E} = 0 \quad \cdots \rho$

$\nu=i$ 成分 $\quad \vec{\nabla}\times\vec{B} - \frac{1}{c}\frac{\partial}{\partial t}\vec{E} = 0 \quad \cdots j$ $\qquad \epsilon^{\mu\alpha\rho\nu}\partial_\alpha F_{\rho\nu} = 0$

↓

および恒等式 $\partial_\alpha(\partial_\rho A_\nu - \partial_\nu A_\rho) + \partial_\rho(\partial_\nu A_\alpha - \partial_\alpha A_\nu) + \partial_\nu(\partial_\alpha A_\rho - \partial_\rho A_\alpha) = 0$

α,ρ,ν の適当な組み合わせを考えると

$$\begin{cases} \vec{\nabla}\times\vec{E} + \frac{1}{c}\frac{\partial}{\partial t}\vec{B} = 0 \\ \vec{\nabla}\cdot\vec{B} = 0 \end{cases}$$

$\vec{E} = (F_{01}, F_{02}, F_{03})$

$\vec{B} = (-F_{23}, -F_{31}, -F_{12})$ $\qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$

ゲージ変換 (gauge transformation)

$A_\mu(x) \longrightarrow A'_\mu(x) = A_\mu(x) + \partial_\mu \omega(x)$

↑ 任意の関数

の下で

$F_{\mu\nu}' \equiv \partial_\mu A'_\nu - \partial_\nu A'_\mu = \partial_\mu(A_\nu + \partial_\nu\omega) - \partial_\nu(A_\mu + \partial_\mu\omega) = F_{\mu\nu}$ gauge invariant

$\Rightarrow$ 作用 $S = -\frac{1}{4}\int F_{\mu\nu}F^{\mu\nu}d^4x$ も不変

一般の $\partial_\mu A^\mu \neq 0$ となる任意の A^μ から出発して $\partial_\mu A'^\mu = \partial_\mu(A^\mu + \partial^\mu\omega) = 0$ とできる

Lorentz ゲージ条件 (Landau ゲージ条件)

∵) $\partial_\mu A^\mu + \partial_\mu\partial^\mu\omega = 0$

$\omega = \frac{-1}{\Box}\partial_\mu A^\mu$

このゲージ条件では Maxwell 方程式は

$\partial_\mu\partial^\mu A^\nu - \partial^\nu\underbrace{\partial_\mu A^\mu}_{0} = \Box A = 0$ 電磁波は光の速さで伝わる

自由粒子

Lagrangian $\quad L = -mc^2\sqrt{1-\left(\frac{v}{c}\right)^2} \sim -mc^2 + \frac{1}{2}mv^2 \quad$ for $\left|\frac{v}{c}\right| \ll 1$

作用 $S = \int L\,dt = -mc^2\int\sqrt{1-\left(\frac{v}{c}\right)^2}\,dt = -mc\int\sqrt{c^2dt^2 - d\vec{x}^2} = -mc\int ds$ Lorentz 不変な線素

運動方程式 $\delta S = \int \delta L\, dt = -mc^2 \int \frac{\frac{1}{c^2}\vec{v}\,\delta\vec{v}}{\sqrt{1-(v/c)^2}}\, dt = -\int \delta\vec{x} \underbrace{\frac{d}{dt}\left(\frac{m\vec{v}}{\sqrt{1-(v/c)^2}}\right)}_{=0} dt = 0$

正準形式

$$\vec{p} = \frac{\delta L}{\delta \dot{\vec{x}}} = mc\,\frac{\frac{1}{c^2}\vec{v}}{\sqrt{1-(v/c)^2}} = \frac{m\vec{v}}{\sqrt{1-(v/c)^2}}$$

$$H = \vec{p}\cdot\dot{\vec{x}} - L = \frac{m\vec{v}^2}{\sqrt{1-(v/c)^2}} + mc^2\sqrt{1-(v/c)^2} = \frac{mc^2}{\sqrt{1-(v/c)^2}} = c\sqrt{\vec{p}^2+(mc)^2}$$

Poisson bracket

$$\{x_k, p_l\}_{PB} = \delta_{kl}$$

$$\frac{d}{dt}p_l = \{H, p_l\}_{PB} = 0$$

$$\frac{d}{dt}x_k = \{H, x_k\}_{PB} = \frac{cp_k}{\sqrt{p^2+(mc)^2}}$$

Lorentz 共変 (5 May 2x)

一般に テンソル $= 0$

$A_{\mu\nu\cdots\lambda} = 0$

の形の方程式は Lorentz 不変(共変) ∵) $A'_{\mu'\nu'\cdots\lambda'} = a^{\mu}_{\mu'} a^{\nu}_{\nu'} \cdots a^{\lambda}_{\lambda'} A_{\mu\nu\cdots\lambda}$

$\frac{d}{dt}H = \{H, H\}_{PB} = 0$ $\quad H = c\sqrt{p^2+m^2c^2} = \frac{mc^2}{\sqrt{1-(v/c)^2}}$ = 全エネルギー

$H^2 - (c\vec{p})^2 = (mc^2)^2$

$p^{\mu} = \left(\frac{1}{c}H, \vec{p}\right) = (p_0, \vec{p})$ 4元運動量 (4-momentum)

$(p^{\mu})^2 \overset{\text{def}}{\equiv} p_0^2 - (\vec{p})^2 = (mc)^2$

Lorentz 不変な制約を与える

(constraint)

$E = \frac{1}{2m}p^2$ (非相対論) $\quad p^0 = E/c + mc$, $E \ll mc^2$

非一量子化 の Schrödinger eq の一般化　　$\rho = \psi^*\psi$　$\vec{j} = \psi^*\vec{\nabla}\psi - (\vec{\nabla}\psi^*)\psi$

$$E = \frac{1}{2m}p^2 = 0 \longrightarrow \left[i\hbar\frac{\partial}{\partial t} - \frac{1}{2m}\left(\frac{\hbar}{i}\vec{\nabla}\right)^2\right]\psi(x) = 0$$

↓

$(p^0)^2 - (\vec{p})^2 = (mc)^2$ において　$p^0 = i\hbar\frac{\partial}{\partial t}$, $\vec{p} = \frac{\hbar}{i}\frac{\partial}{\partial \vec{x}}$　$p_\mu = i\hbar\frac{\partial}{\partial x^\mu}$

$$\left[(p_\mu)^2 - (mc)^2\right]\phi(x) = \left[\left(i\hbar\frac{\partial}{\partial x^\mu}\right)^2 - (mc)^2\right]\phi(x) = 0$$

あるいは

$$\left[\left(\frac{\partial}{\partial x^\mu}\right)^2 + \left(\frac{mc}{\hbar}\right)^2\right]\phi(x) = 0$$　Klein-Gordon 方程式（波動場）

この方程式を与える Lagrangian (density)

$$S = \int \mathcal{L}\, d^4x = \int \left[\partial_\mu\phi^*(x)\partial^\mu\phi(x) - \left(\frac{mc}{\hbar}\right)^2\phi^*(x)\phi(x)\right]d^4x$$

$$= \int L\, dt = \int\left[(\partial_0\phi^*)(\partial_0\phi) - \frac{\partial}{\partial\vec{x}}\phi^*\frac{\partial}{\partial\vec{x}}\phi - \left(\frac{mc}{\hbar}\right)^2\phi^*\phi\right]d^4x$$

作用 S は Lorentz 変換 $\begin{cases} x'^\mu = a^\mu{}_\nu x^\nu \text{ とすると不変} & x' = Ax \\ \phi'(x') = \phi(x) \cdots \text{スカラー場} & \phi'(x) = \phi(A^{-1}x) \end{cases}$

$d^4x' = d^4x$

$$S' = \int \mathcal{L}(\partial'_\mu\phi'(x), \phi'(x)\, d^4x$$

$$= \int \mathcal{L}(\partial^x_\mu\phi(A^{-1}x), \phi(A^{-1}x))\, d^4x \quad (d^4(A^{-1}x) = d^4y)$$

$$= \int \mathcal{L}(\partial^y_\mu\phi(y), \phi(y))\, d^4y = S$$

対称性（Symmetry）と保存則（conservation law）

作用 S

$$S = \int\left[\partial_\mu\phi^*\partial^\mu\phi - \left(\frac{mc}{\hbar}\right)^2\phi^*\phi\right]d^4x$$

α を定数として変換（位相変換）　$\phi(x) = e^{i\alpha}\phi(x)$: $\phi^*(x) = e^{-i\alpha}\phi^*(x)$

の下で不変

この時 S は対称性をもつという

⇒ α を座標に依存する $\alpha(x)$ に一般化する

$$S' \equiv \int \left[\partial_\mu \left(e^{-i\alpha(x)} \phi^*(x) \right) \partial^\mu \left(e^{i\alpha(x)} \phi(x) \right) - \left(\frac{mc}{\hbar} \right)^2 \phi^*(x) \underbrace{e^{-i\alpha(x)} e^{i\alpha(x)}}_{1} \phi(x) \right] d^4x$$

$$= S - \int (\partial_\mu \alpha(x)) \, i \left[\phi^*(x) \partial^\mu \phi(x) - \partial^\mu \phi^*(x) \phi(x) \right] d^4x \qquad (\alpha(x) \text{ の 1 次の精度で})$$

$\partial_\mu \alpha(x)$ の係数として Noether Current を定義する.

$$J^\mu(x) \equiv i \left[\phi^*(x) \partial^\mu \phi(x) - \partial^\mu \phi^*(x) \phi(x) \right]$$

Noether の定理

このカレントは保存する $\partial_\mu J^\mu(x) \equiv \frac{1}{c} \frac{\partial}{\partial t} J^0 + \frac{\partial}{\partial x^k} J^k = 0$

但、運動方程式を使う

$\because$) $\partial_\mu J^\mu = i \left[\phi^*(x) \partial_\mu \partial^\mu \phi(x) - \partial_\mu \partial^\mu \phi^*(x) \phi(x) \right]$ $\qquad \partial_\mu \partial^\mu \phi + \left(\frac{mc}{\hbar} \right)^2 \phi = 0$

K-G eq

$$= i \left[- \phi^*(x) \left(\frac{mc}{\hbar} \right)^2 \phi(x) - \left(\frac{mc}{\hbar} \right)^2 \phi^*(x) \phi(x) \right] = 0$$

このとき

$$\frac{1}{c} \frac{d}{dt} \int J^0(t, \vec{x}) d^3x = \int \frac{1}{c} \frac{\partial}{\partial t} J^0(t, \vec{x}) d^3x = - \int \partial_k J^k d^3x = - \int_{\text{表面}} J^k dS_k = 0$$

すなわち

$Q = \int J^0(t, \vec{x}) d^3x$ は 時間的に変わらない = 保存量

Noether Current

$$J^\mu(x) = i \left(\phi^*(x) \partial^0 \phi(x) - \partial^0 \phi^* \phi(x), \; \phi^*(x) \partial^k \phi(x) - \partial^k \phi^*(x) \phi(x) \right)$$

は Schrödinger 方程式の

$$\left(\psi^*(x) \psi(x), \; \frac{-i\hbar}{2m} \left(\psi^*(x) \vec{\nabla} \psi(x) - \nabla \psi^*(x) \cdot \psi(x) \right) \right)$$

$\parallel$

ρ 確率密度 の一般化 と考えられる

従って 素朴には

$J^0(x) = i\left[\phi^*(x)\frac{\partial\phi(x)}{c\partial t} - \frac{\partial}{c\partial t}\phi^*(x)\phi(x)\right]$ が 確率密度 の一般化と考えられる.

運動方程式

$$\delta S = \int\left[\partial_\mu \delta\phi^* \partial^\mu\phi - \left(\frac{mc}{\hbar}\right)^2 \delta\phi^*\phi\right]d^4x = 0$$

$$\partial_\mu\partial^\mu\phi(x) + \left(\frac{mc}{\hbar}\right)^2\phi(x) = 0$$

自由場の解 $\phi(x) = Ne^{-i(\omega t - \vec{k}\vec{x})/\hbar}$ を代入

$$\left[\left(\frac{\omega}{c}\right)^2 - \vec{k}^2 - \left(\frac{mc}{\hbar}\right)^2\right]\phi(x) = 0$$

$$\left(\frac{\omega}{c}\right)^2 = \vec{k}^2 + \left(\frac{mc}{\hbar}\right)^2 \longrightarrow \omega = \pm c\sqrt{(\vec{k})^2 + (mc)^2} = \pm E(k) \qquad E(k) > 0$$

$\phi(x) = Ne^{-i(Et - \vec{k}\cdot\vec{x})/\hbar}$ 正エネルギー解

$= Ne^{i(Et - \vec{k}\cdot\vec{x})/\hbar}$ 負 〃

∵) $i\hbar\frac{\partial}{\partial t}\phi(x) = \pm E\,\phi(x)$

このとき $J^0(x) = 2|N|^2(\omega/\hbar) \gtrless 0$ ⇒ $J^0(x)$ は確率密度と見なせない

⇒ $J^0(x)$ は 場の理論の枠内では 電荷密度と考えられる.

H.W. (大学院) (Pauli-Weisskopf)

$\omega = E = c\sqrt{(\vec{k})^2 + (mc)^2} > 0$ に限定すると

$i\hbar\frac{\partial}{\partial t}\phi(x) = c\sqrt{\left(\frac{\hbar}{i}\frac{\partial}{\partial x}\right)^2 + (mc)^2}\,\phi(x)$ となるが何か問題か?

Newton-Wigner Rev. Mod. Phys. 21 (49) 400

場の量子論 (← 第2量子化)

例) 実のスカラー場 A_μ 実ベクトル場

$\left[\Box + \left(\frac{mc}{\hbar}\right)^2\right]\phi(x) = 0$ $\phi^* = \phi$ ← 実 (real field)

スカラー ← スピン 0

Lorentz 変換則

$$\begin{cases} x'^\mu = a^\mu{}_\nu x^\nu \\ \phi'(x') = \phi(x) \quad \leftarrow \text{スピン } 0 \end{cases}$$

$$\left(A'^\mu(x') = a^\mu{}_\nu A^\nu(x) \right)$$

$\psi = \frac{1}{\sqrt{2}}(\psi_{re} + i\psi_{im})$ の下で作用

$$S = \frac{1}{2}\int \left[\partial_\mu \phi(x) \partial^\mu \phi(x) - \left(\frac{mc}{\hbar}\right)^2 \phi(x)\phi(x) \right] d^4x \quad \text{は不変}$$

作用の変分 $\delta S = \int \left[\partial_\mu \delta\phi \partial^\mu \phi - \left(\frac{mc}{\hbar}\right)^2 \delta\phi \phi \right] d^4x$

$$= \int \delta\phi \left[-\partial_\mu \partial^\mu \phi - \left(\frac{mc}{\hbar}\right)^2 \phi \right] d^4x = 0$$

$$\rightarrow \partial_\mu \partial^\mu \phi + \left(\frac{mc}{\hbar}\right)^2 \phi = \left[\left(\frac{\partial}{c\partial t}\right)^2 - \left(\frac{\partial}{\partial x}\right)^2 + \left(\frac{mc}{\hbar}\right)^2 \right] \phi(x) = 0$$

古典的な4次元時空中の場の方程式

$\Updownarrow$

Maxwell 方程式

量子化 $\{\varphi_\ell(\vec{x})\}$ 実の正規直交系 $\int \varphi_\ell(\vec{x}) \varphi_m(\vec{x}) d^3x = \delta_{\ell m}$

$=$ t によらない

$$\phi(t, x) = \sum_\ell q_\ell(t) \varphi_\ell(\vec{x})$$

$$L = \int \mathcal{L}(\partial_\mu \phi, \phi) d^3x = \frac{1}{2}\sum_\ell \frac{1}{c^2}(\dot{q}_\ell)^2 - \frac{1}{2}\sum_{\ell,n} \omega_{\ell n} q_\ell q_n$$

但. $\omega_{\ell n} = \int \left(\sum_{k=1}^{3} \frac{\partial}{\partial x^k} \varphi_\ell \frac{\partial}{\partial x^k} \varphi_n + \left(\frac{mc}{\hbar}\right)^2 \varphi_\ell \varphi_n \right) d^3x$

$\Rightarrow$ 無限個の振動子

正準理論

$$p_\ell = \frac{\partial L}{\partial \dot{q}_\ell} = \frac{1}{c^2}\dot{q}_\ell \qquad H = \sum_\ell p_\ell \dot{q}_\ell - L = \frac{1}{2}\sum_\ell c^2 p_\ell^2 + \frac{1}{2}\sum_{\ell,n} \omega_{\ell n} q_\ell q_n$$

量子化

$$[\hat{p}_\ell, \hat{q}_n] = -i\hbar \delta_{\ell n}$$

$\downarrow$ $\varphi_\ell(\vec{x})$ と $\varphi_n(\vec{y})$ を掛けて ℓ, n に関して和をとる

$$[\hat{\pi}_\phi(t, \vec{x}), \hat{\phi}(t, \vec{y})] = -i\hbar \delta^3(\vec{x} - \vec{y})$$

宿題 6/3 レポート出題

宿題 6/4 でいしゅつ

但.

$$\begin{cases} \hat{\pi}_\phi(t,\vec{x}) = \sum_k \hat{P}_k(t)\,\varphi_k(\vec{x}) = \frac{1}{c^2}\dot{\hat{\phi}}(t,\vec{x}) \\ \sum_k \varphi_k(\vec{x})\,\varphi_k(\vec{y}) = \delta^3(\vec{x}-\vec{y}) \end{cases}$$

場の理論としてやると

$$\pi_\phi(t,\vec{x}) = \frac{\delta}{\delta\dot{\phi}(t,\vec{x})} S = \frac{\delta}{\delta\dot{\phi}(t,\vec{x})}\int \mathcal{L}(\partial_\mu\phi,\phi)\,d^4x = \frac{1}{c^2}\dot{\phi}(t,\vec{x})$$

▷ Jun 3

関数微分

$$\frac{\delta}{\delta f(t')}\int f(t)\,dt = 1 \qquad \text{具体的には} \quad \frac{\delta}{\delta(\Delta t\, f(t_n))}\left(\sum_n f(t_n)\Delta t\right) = 1$$

Lesbegee 積分

このとき $\dfrac{\delta}{\delta f(t')}\int \frac{1}{2} f(t)^2\,dt = f(t')$

∵) $\dfrac{\delta}{\delta(\Delta t\, f(t_n))}\sum_n \frac{1}{2}(f(t_n)\Delta t)^2\frac{1}{\Delta t} = f(t_{n'})\Delta t\left(\frac{1}{\Delta t}\right) = f(t_{n'})$

一般に $\dfrac{\delta}{\delta f(t')}\int f(t)^n dt = n f(t')^{n-1}$ ∵) 各自

このとき

$$\int \mathcal{D}f\,\frac{\delta}{\delta f(t')}\left(e^{-\frac{1}{2}\int f(t)^2 dt}\right) = \int \mathcal{D}f\,(-f(t'))\,e^{-\frac{1}{2}\int f(t)^2 dt}$$

$$\propto \int \underbrace{\prod_{n=1}^{\infty} df(t_n)}_{\mathcal{D}f}\left[-f(t')\,e^{-\frac{1}{2}\sum_m f(t_m)^2\Delta t}\right] = 0 \quad \Leftrightarrow \text{経路積分で運動方程式}$$

$$\left(\text{c.f. } \int_{-\infty}^{\infty} dx\,\frac{dg(x)}{dx} = g(+\infty) - g(-\infty) = 0\right)$$

場の理論

$$\pi_\phi(t,\vec{x}) = \frac{\delta}{\delta\dot{\phi}(t,\vec{x})}\int \mathcal{L}(\partial_\mu\phi,\phi)\,d^4x$$

$$= \frac{1}{c^2}\dot{\phi}(t,\vec{x})$$

$$\begin{cases} p = \dfrac{\delta}{\delta\dot{q}} L = m\dot{q} \\ p = \dfrac{\delta}{\delta\dot{q}}\int L\,dt \end{cases}$$

Hamiltonian

$$\mathcal{H} = \int d^3x\left[\pi_\phi(t,\vec{x})\,\dot{\phi}(t,\vec{x}) - \mathcal{L}\right] = \frac{1}{2}\int d^3x\left[c^2\pi_\phi(t,\vec{x})^2 + (\partial_k\phi(t,\vec{x}))^2 + \left(\frac{mc}{\hbar}\right)^2\phi(t,\vec{x})^2\right]$$

但し $\mathcal{L} = \frac{1}{2}\partial_\mu\phi(x)\,\partial^\mu\phi(x) - \frac{1}{2}\left(\frac{mc}{\hbar}\right)^2\phi(x)^2$

$= \frac{1}{2c^2}\dot{\phi}(x)\dot{\phi}(x) - \frac{1}{2}\sum_{k=1}^{3}\partial_k\phi(x)\,\partial_k\phi(x) - \frac{1}{2}\left(\frac{mc}{\hbar}\right)^2\phi(x)^2$

場の量子化

$\left[\hat{\pi}_\phi(t,\vec{x}),\ \hat{\phi}(t,\vec{y})\right] = -i\hbar\,\delta^3(\vec{x}-\vec{y})$; $\hat{\pi}_\phi(t,\vec{x}) = \frac{1}{c^2}\dot{\hat{\phi}}(t,\vec{x})$

$\left[\hat{\pi}_\phi(t,\vec{x}),\ \hat{\pi}_\phi(t,\vec{y})\right] = \left[\hat{\phi}(t,\vec{x}),\ \hat{\phi}(t,\vec{y})\right] = 0$

$\hat{H} = \frac{1}{2}\int d^3x\left[c^2\hat{\pi}_\phi(t,\vec{x})^2 + \left(\partial_k\hat{\phi}(t,\vec{x})\right)^2 + \left(\frac{mc}{\hbar}\right)^2\hat{\phi}(t,\vec{x})^2\right]$

正準方程式 (Canonical eq. of motion)

$$\begin{cases} i\hbar\frac{\partial}{\partial t}\hat{\phi}(t,\vec{x}) = \left[\hat{\phi}(t,\vec{x}),\ \hat{H}\right] = i\hbar c^2\hat{\pi}_\phi(t,\vec{x}) \\ i\hbar\frac{\partial}{\partial t}\hat{\pi}_\phi(t,\vec{x}) = \left[\hat{\pi}_\phi(t,\vec{x}),\ \hat{H}\right] = i\hbar\left[\sum_{k=1}^{3}\left(\frac{\partial}{\partial x_k}\right)^2\hat{\phi}(t,\vec{x}) - \left(\frac{mc}{\hbar}\right)^2\hat{\phi}(t,\vec{x})\right] \end{cases}$$

$$\Rightarrow \frac{1}{c^2}\ddot{\hat{\phi}}(t,\vec{x}) - \left(\frac{\partial}{\partial x_k}\right)^2\hat{\phi}(t,\vec{x}) + \left(\frac{mc}{\hbar}\right)^2\hat{\phi}(t,\vec{x}) = \left(\Box + \left(\frac{mc}{\hbar}\right)^2\right)\hat{\phi} = 0$$

$\left(\Box + \left(\frac{mc}{\hbar}\right)^2\right)f(t,\vec{x}) = 0 \rightarrow f(t,\vec{x}) \propto e^{\pm i(Et - \vec{k}\cdot\vec{x})/\hbar}$ $\quad E = c\sqrt{(k)^2 + (mc)^2}$

平面波の完全系で展開すると

$$\hat{\phi}(x) = \int \frac{d^3k}{(2\pi\hbar)^{3/2}}\frac{\hbar c}{\sqrt{2E}}\left[\hat{a}_{\vec{k}}\,e^{-i(Et-\vec{k}\vec{x})/\hbar} + \hat{a}^\dagger_{\vec{k}}\,e^{i(Et-\vec{k}\cdot\vec{x})/\hbar}\right]$$

↑ 正の振動数　　↑ 負 ‥　　∵) $H = i\hbar\frac{\partial}{\partial t}$

$\hat{\phi}(x)^\dagger = \hat{\phi}(x)$ ← 実の場

交換関係

$\left[\dot{\hat{\phi}}(t,\vec{x}),\ \hat{\phi}(t,\vec{y})\right] =$

$$= \frac{(\hbar c)^2}{(2\pi\hbar)^3}\int\frac{d^3k}{\sqrt{2E}}\frac{d^3k'}{\sqrt{2E}}\left[-iE\,\hat{a}_k e^{-i(Et-kx)/\hbar} + iE\,\hat{a}^\dagger_k e^{i(Et-kx)/\hbar},\ \hat{a}_{k'}e^{-i(E't-k'y)/\hbar} + \hat{a}^\dagger_{k'}e^{i(E't-k'y)/\hbar}\right]$$

$E(k) = c\sqrt{(\vec{k})^2 + (mc)^2}$

$$= \frac{(\hbar c)^2}{(2\pi\hbar)^3} \int \frac{d^3k}{\sqrt{2E}} \frac{d^3k'}{\sqrt{2E'}} \left\{ \left[-\frac{iE}{\hbar} \hat{a}_{\vec{k}} , \hat{a}^{+}_{\vec{k}'} \right] e^{-i(Et - \vec{k}\vec{x})/\hbar} e^{i(E't - \vec{k}'\vec{y})/\hbar} \right.$$

$$\left. + \left[\frac{iE}{\hbar} \hat{a}^{+}_{\vec{k}} , \hat{a}_{\vec{k}'} \right] e^{i(Et - \vec{k}\vec{x})/\hbar} e^{-i(E't - \vec{k}'\vec{y})/\hbar} \right\}$$

$$= \frac{(\hbar c)^2}{(2\pi\hbar)^3} \int \frac{d^3k\, d^3k'}{2E} \left[-\frac{iE}{\hbar} \delta^3(\vec{k} - \vec{k}') e^{i\vec{k}(\vec{x} - \vec{y})/\hbar} - \frac{iE}{\hbar} \delta^3(\vec{k} - \vec{k}') e^{-i\vec{k}(\vec{x} - \vec{y})/\hbar} \right]$$

$$= \frac{(\hbar c)^2}{(2\pi\hbar)^3} \left(\frac{-2i}{2\hbar} \right) (2\pi)^3 \delta\left(\frac{\vec{x} - \vec{y}}{\hbar} \right) = -i\hbar c^2 \delta^3(\vec{x} - \vec{y})$$

使った条件 $[\hat{a}_{\vec{k}}, \hat{a}_{\vec{k}'}] = [\hat{a}^{+}_{\vec{k}}, \hat{a}^{+}_{\vec{k}'}] = 0$, $[\hat{a}_{\vec{k}}, \hat{a}^{+}_{\vec{k}'}] = \delta^3(\vec{k} - \vec{k}')$

生成・消滅演算子のレベルでの量子化

次に、

$$\frac{1}{2} \int d^3x \frac{\partial}{c\partial t} \hat{\phi}(t, \vec{x}) \frac{\partial}{c\partial t} \hat{\phi}(t, \vec{x}) = \quad \phi \sim \int \frac{d^3k}{\sqrt{2E}} \left[\hat{a}_{\vec{k}'} e^{-i(Et - \vec{k}'\vec{x})/\hbar} + \hat{a}^{+}_{\vec{k}'} e^{i(Et - \vec{k}'\vec{x})/\hbar} \right]$$

$$= \frac{1}{2} \int d^3k \left[\frac{1}{2} E(\vec{k}) \left(\hat{a}_{\vec{k}} \hat{a}^{+}_{\vec{k}} + \hat{a}^{+}_{\vec{k}} \hat{a}_{\vec{k}} \right) - \frac{1}{2} E(\vec{k}) \hat{a}_{\vec{k}} \hat{a}_{-\vec{k}} e^{-2iE(\vec{k})t/\hbar} - \frac{1}{2} E(\vec{k}) \hat{a}^{+}_{\vec{k}} \hat{a}^{+}_{-\vec{k}} e^{2iE(\vec{k})t/\hbar} \right]$$

および

$$\frac{1}{2} \int d^3x \left[\sum_{l=1}^{3} \frac{\partial}{\partial x_l} \hat{\phi}(t, \vec{x}) \frac{\partial}{\partial l} \hat{\phi}(t, \vec{x}) + \left(\frac{mc}{\hbar} \right)^2 \hat{\phi}(t, \vec{x}) \hat{\phi}(t, \vec{x}) \right]$$

$$= \frac{1}{2} \int d^3k \left[\frac{1}{2} E(\vec{k}) \left(\hat{a}_{\vec{k}} \hat{a}^{+}_{\vec{k}} + \hat{a}^{+}_{\vec{k}} \hat{a}_{\vec{k}} \right) + \frac{1}{2} E(\vec{k}) \hat{a}_{\vec{k}} \hat{a}_{-\vec{k}} e^{-2iE(\vec{k})t/\hbar} + \frac{1}{2} E(\vec{k}) \hat{a}^{+}_{\vec{k}} \hat{a}^{+}_{-\vec{k}} e^{2iE(\vec{k})t/\hbar} \right]$$

$$\hat{H} = \int d^3k \, \frac{1}{2} E(\vec{k}) \left[\hat{a}_{\vec{k}} \hat{a}^{+}_{\vec{k}} + \hat{a}^{+}_{\vec{k}} \hat{a}_{\vec{k}} \right]$$

(i) Casimir $-\frac{\partial}{\partial l} \circ E(l) = \vec{f}$

(ii) Cavity QED

$$= \int d^3k \, E(\vec{k}) \left[\hat{a}^{+}_{\vec{k}} \hat{a}_{\vec{k}} + \frac{1}{2} \right] \geq 0$$

$= 0$ 点エネルギー

正規順序化 (normal ordering)

$$:\hat{H}: = \int d^3k \, E(\vec{k}) \hat{a}^{+}_{\vec{k}} \hat{a}_{\vec{k}}$$

- 消滅演算子は右
- 生成 〃 左
- 定数は無視

$\hbar = \frac{200\,\mathrm{MeV\cdot fm}}{3\cdot 10^{8}\,\mathrm{m/s}} = 10^{3}\cdot 10^{-15}\cdot 10^{-8}\,\mathrm{MeV\cdot s}$

$10^{-21}\,\mathrm{m\cdot V\cdot s} = 10^{-15}\,\mathrm{eV\cdot s}$

[Report]

1. $\Delta(x-y) \equiv [\hat{\phi}(x), \hat{\phi}(y)]$ を計算し（但、一般に $x^0 \neq y^0$ とする）

次のことを示せ

(i) $\Delta(x-y)$ は Lorentz 不変

(ii) $\Delta(x-y) = 0$ for $(x-y)^2 = (x^0-y^0)^2 - (\vec{x}-\vec{y})^2 < 0$ ← 因果律（Causality）

(iii) $\left.\frac{\partial}{\partial x^0}\Delta(x-y)\right|_{x^0=y^0} = -ihc\,\delta^3(\vec{x}-\vec{y})$

2. 自由なスカラー場に対する $\hat{N}$ の計算を check せよ.

3. 4元ベクトル $J^\mu(x)$ $\partial_\mu J^\mu(x) = 0$ なら

(i) $\frac{d}{dt}\int J^0(t,\vec{x})\,d^3x = 0$

(ii) $Q = \int J^0(t,\vec{x})\,d^3x$ は Lorentz 不変

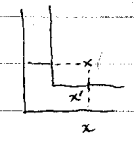
以上 さういふつまり16/17か.

場の理論におけるエネルギー運動量テンソル

例、実スカラー場

$S = \frac{1}{2}\int\left[\partial_\mu\phi(x)\,\partial^\mu\phi(x) - \left(\frac{mc}{\hbar}\right)^2\phi(x)\phi(x)\right]d^4x$ は座標の並進 $x^\mu \to x^{\mu\prime} = x^\mu + \varepsilon^\mu$ （ε^μ は定数）に対して不変

∵) $\phi'(x') = \phi(x)$ （定義）



$\left.\begin{aligned}\phi'(x+\varepsilon) &= \phi(x)\\ \phi'(x) &= \phi(x-\varepsilon)\end{aligned}\right\}$ ↘

$S' = \frac{1}{2}\int\left[\partial_\mu\phi'(x)\,\partial^\mu\phi'(x) - \left(\frac{mc}{\hbar}\right)^2\phi'(x)\,\phi'(x)\right]d^4x$

$= \frac{1}{2}\int\left[\partial_\mu\phi(x-\varepsilon)\,\partial^\mu\phi(x-\varepsilon) - \left(\frac{mc}{\hbar}\right)^2\phi(x-\varepsilon)\,\phi(x-\varepsilon)\right]d^4x$

$= \frac{1}{2}\int\left[\partial''_\mu\phi(x'')\,\partial''^\mu\phi(x'') - \left(\frac{mc}{\hbar}\right)^2\phi(x'')\,\phi(x'')\right]d^4x'' = S$

Neother Current （無限小）

（ε^μ を x に依存する $\varepsilon^\mu(x)$ に変える）

$S' = \frac{1}{2}\int\left[\partial_\mu\phi(x^\nu - \varepsilon^\nu(x))\,\partial^\mu\phi(x^\nu - \varepsilon^\nu(x)) - \left(\frac{mc}{\hbar}\right)^2\phi(x^\nu-\varepsilon^\nu(x))\,\phi(x^\nu-\varepsilon^\nu(x))\right]d^4x = S$

$-\int d^4x\left\{\partial_\mu\varepsilon^\nu(x)\left[\partial_\nu\phi(x)\,\partial^\mu\phi(x)\right] - (\partial_\nu\varepsilon^\nu(x))\left[\frac{1}{2}\partial_\mu\phi(x)\,\partial^\mu\phi(x) - \frac{1}{2}\left(\frac{mc}{\hbar}\right)^2\phi^2(x)\right]\right\}$

$\partial_\mu\varepsilon^\nu(x)\,T^\mu{}_\nu(x)$

並進 $x^\mu \to x^\mu + \varepsilon^\mu$ ▷ Jun 17

$\downarrow$ $\quad T_{\mu\nu} = g_{\nu\alpha} T_\mu{}^\alpha$

Energy - Momentum Tensor

$T^\nu_\mu(x) \to \partial_\nu T^\nu_\mu(x) = 0$ $\quad | \quad T_{\mu\nu} = T_{\nu\mu}$ (Lorentz 不変性)

$P_\mu = \int T^0_\mu(x)\, d^3x$

$\frac{d}{dt} P_\mu = 0$

場の量子化 $\hat{P}_\mu = \hat{H} = \int d^3k\, E(\vec{k})\, \hat{n}_{\vec{k}}$ $\quad E(\vec{k}) = c\sqrt{(\vec{k})^2 + (mc)^2}$ $\quad \hat{n}_{\vec{k}} = \hat{a}^+_{\vec{k}} \hat{a}_{\vec{k}}$ 粒子数

$\hat{P}_e/c = \int d^3k\, \vec{k}\, \hat{n}_{\vec{k}}$

Fock 状態

$|\vec{k}\rangle \equiv \hat{a}^+_{\vec{k}} |0\rangle$

$\hat{H}|\vec{k}\rangle = E(\vec{k})|\vec{k}\rangle$

$\vec{P}/c\,|\vec{k}\rangle = \vec{k}\,|\vec{k}\rangle$

場 $\rightleftharpoons$ 粒子

量子化

Schrödinger eq

$i\hbar \frac{\partial}{\partial t} |\Psi(t)\rangle = \hat{H} |\Psi(t)\rangle$

$\hat{H}^+ = H \to$ 確率の保存

Feynman Propagator $\quad G(\vec{x}, x^0; \vec{y}, y^0)$

量子力学 $\sum_n \varphi_n(\vec{x})\, \varphi_n(\vec{y})\, e^{-iE_n(x^0 - y^0)/\hbar c}$ $\quad$ for $x^0 > y^0$

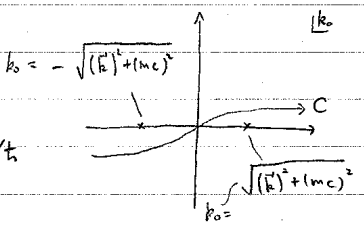
$\Psi(\vec{x}, x^0) = \int G(\vec{x}, x^0; \vec{y}, y^0)\, \Psi(\vec{y}, y^0)\, d^3y$

$i G_F(x-y) = \langle 0 | T\, \hat{\phi}(x) \hat{\phi}(y) | 0 \rangle$

↖ Time Ordering

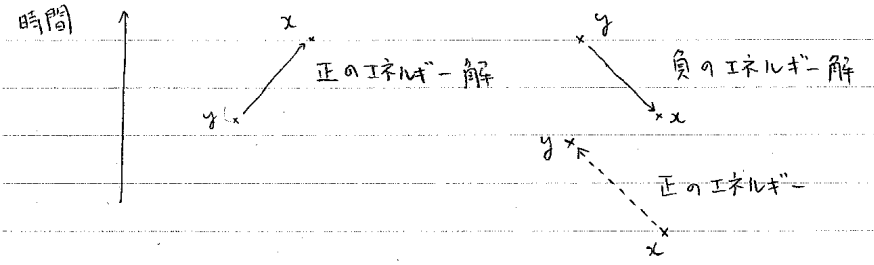
$= \int_C \frac{d^4k}{(2\pi\hbar)^4} \cdot \frac{ihc}{(k/\hbar)^2 - (mc/\hbar)^2 + i\varepsilon} e^{-ik_\mu (x-y)^\mu/\hbar}$

↖ Feynman's ε



正のエネルギー解 → $x^0 > y^0$ 積分路は下に閉じる

負 〃 → $y^0 > x^0$ 〃 上



複素スカラー場　　スカラー → スピン0

$$\begin{cases} \text{電荷} \\ \text{反粒子} \end{cases} \quad \mathcal{L} = \partial_\mu \varphi(x)\, \partial^\mu \varphi(x)^* - \left(\frac{mc}{\hbar}\right)^2 \varphi(x)\,\varphi(x)^*$$

$$= \frac{1}{2}\left[\partial_\mu \phi^{(1)}(x)\, \partial^\mu \phi^{(1)}(x) - \left(\frac{mc}{\hbar}\right)^2 \phi^{(1)}(x)\,\phi^{(1)}(x)\right] \qquad \varphi(x) = \frac{1}{\sqrt{2}}\left(\phi^{(1)}(x) + i\,\phi^{(2)}(x)\right)$$

Hilbert空間の計量

$$+ \frac{1}{2}\left[\partial_\mu \phi^{(2)}(x)\, \partial^\mu \phi^{(2)}(x) - \left(\frac{mc}{\hbar}\right)^2 \phi^{(2)}(x)\,\phi^{(2)}(x)\right]$$

量子化　実スカラー場の結果を用いて

$$\hat{\phi}^{(\lambda)}(x) = \int \frac{d^3k\,\hbar c}{\sqrt{(2\pi\hbar)^3 2E}} \left[\hat{a}^{(\lambda)}_{\vec{k}}\, e^{-i(Et - \vec{k}\cdot\vec{x})/\hbar} + \hat{a}^{+}_{\vec{k}}\, e^{i(Et - \vec{k}\cdot\vec{x})/\hbar}\right]$$

↑ $\lambda = 1, 2$

量子化条件 $\left[\hat{a}^{(\lambda)}_{\vec{k}},\ \hat{a}^{+(n)}_{\vec{k}'}\right] = \delta^3(\vec{k}-\vec{k}')\,\delta_{\lambda n}$　他は全て0

ここで改めて

$$\begin{cases} \hat{a}_{\vec{k}} = \frac{1}{\sqrt{2}}\left(\hat{a}^{(1)}_{\vec{k}} + i\,\hat{a}^{(2)}_{\vec{k}}\right) & \left[\hat{a}_{\vec{k}},\ \hat{a}^{+}_{\vec{k}'}\right] = \delta^3(\vec{k}-\vec{k}') \\ \hat{b}_{\vec{k}} = \frac{1}{\sqrt{2}}\left(\hat{a}^{(1)}_{\vec{k}} - i\,\hat{a}^{(2)}_{\vec{k}}\right) & \left[\hat{b}_{\vec{k}},\ \hat{b}^{+}_{\vec{k}'}\right] = \delta^3(\vec{k}-\vec{k}') \quad \text{他も全て0} \end{cases} \Longrightarrow$$

$\mathcal{L}$ は2つのスカラー場の足し算なので

$$\hat{H} = \int d^3k\, E(\vec{k})\left[\hat{a}^{(1)+}_{\vec{k}}\,\hat{a}^{(1)}_{\vec{k}} + \hat{a}^{(2)+}_{\vec{k}}\,\hat{a}^{(2)}_{\vec{k}}\right] = \int d^3k\, E(\vec{k})\left[\hat{a}^{+}_{\vec{k}}\,\hat{a}_{\vec{k}} + \hat{b}^{+}_{\vec{k}}\,\hat{b}_{\vec{k}}\right]$$

$+1$ ← normal ordering

$$\vec{P}/c = \int d^3k\,\vec{k}\,[\quad\text{〃}\quad] = \int d^3k\,\vec{k}\,[\quad\text{〃}\quad]$$

除ける

電流の演算子

ゲージ変換 $\begin{cases} \varphi(x) \to e^{i\alpha}\varphi(x) & \alpha = \text{定数} \\ \varphi^*(x) \to e^{-i\alpha}\varphi^*(x) \end{cases}$

に関する Neother Current

$$J^\mu(x) = e\left(\frac{1}{\hbar c}\right)\left[\varphi^*(x)\,\partial^\mu\varphi(x) - \varphi(x)\,\partial^\mu\varphi^*(x)\right]$$

$$= \left(\frac{e}{\hbar c}\right)\left[\phi^{(2)}(x)\,\partial^\mu\phi^{(1)}(x) - \partial^\mu\phi^{(2)}(x)\,\phi^{(1)}(x)\right]$$

$\partial_\mu J^\mu = 0$ 保存流

電荷 (charge)

$$Q \equiv \int d^3x\, J^0(t,\vec{x}) = ie\int d^3k\left[\hat{a}^{(1)\dagger}_{\vec{k}}\hat{a}^{(2)}_{\vec{k}} - \hat{a}^{(2)\dagger}_{\vec{k}}\hat{a}^{(1)}_{\vec{k}}\right] = e\int d^3k\left[\hat{a}^\dagger_{\vec{k}}\hat{a}_{\vec{k}} - \hat{b}^\dagger_{\vec{k}}\hat{b}_{\vec{k}}\right]$$

Fock States

真空 $|0\rangle$ (normalizable $\langle 0|0\rangle = 1$)

$\hat{a}_{\vec{k}}|0\rangle = \hat{b}_{\vec{k}}|0\rangle = 0$

$\hat{H}|0\rangle = 0$, $Q|0\rangle = 0$, $\hat{P}/c|0\rangle = 0$

一粒子状態

$\hat{a}^\dagger_{\vec{k}}|0\rangle$, $\hat{b}^\dagger_{\vec{k}}|0\rangle$

$c\sqrt{(\vec{k})^2+(mc)^2}$

$\hat{H}(\hat{a}^\dagger_{\vec{k}}\|0\rangle) = E(\vec{k})(\hat{a}^\dagger_{\vec{k}}\|0\rangle)$	$\hat{H}(\hat{b}^\dagger_{\vec{k}}\|0\rangle) = E(\vec{k})(\hat{b}^\dagger_{\vec{k}}\|0\rangle)$
$\hat{P}/c(\cdots) = \vec{k}(\cdots)$	$\hat{P}/c(\cdots) = \vec{k}(\cdots)$
$Q(\cdots) = e(\cdots)$	$Q(\cdots) = -e(\cdots)$

まとめると

	粒子	反粒子
一粒子状態	$\hat{a}^\dagger_{\vec{k}}\|0\rangle$	$\hat{b}^\dagger_{\vec{k}}\|0\rangle$
エネルギー H	$c\sqrt{(\vec{k})^2+(mc)^2}$	$c\sqrt{(\vec{k})^2+(mc)^2}$
運動量 $\vec{P}/c$	$\vec{k}$	$\vec{k}$
電荷 Q	e	$-e$

Feynman 伝搬関数

$$\langle 0|T\hat{\varphi}(x)\hat{\varphi}(y)^{+}|0\rangle = \frac{1}{2}\left[\langle 0|T\hat{\phi}^{(1)}(x)\hat{\phi}^{(1)}(y)|0\rangle + \langle 0|T\hat{\phi}^{(2)}(x)\hat{\phi}^{(2)}(y)|0\rangle\right]$$

$$= \int_C \frac{d^4k}{(2\pi\hbar)^4}\cdot\frac{i\hbar c}{(k_\mu/\hbar)^2-(mc/\hbar)^2+i\varepsilon}e^{-ik_\mu(x-y)^\mu/\hbar}$$

電荷の保存と関係する

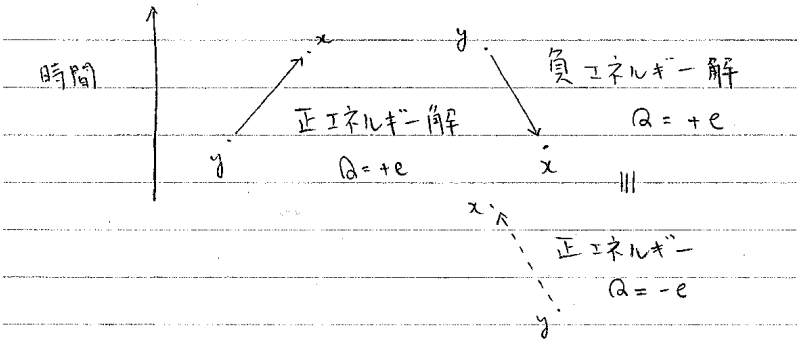
$$\langle 0|T\hat{\varphi}(x)\hat{\varphi}(y)|0\rangle = \frac{1}{2}\left[\langle 0|T\hat{\phi}^{(1)}(x)\hat{\phi}^{(1)}(y)|0\rangle - \langle 0|T\hat{\phi}^{(2)}(x)\hat{\phi}^{(2)}(y)|0\rangle\right] = 0 = \langle 0|T\hat{\varphi}(x)^{+}\hat{\varphi}^{+}(y)|0\rangle$$

$$\hat{\varphi}(x) \to e^{i\alpha}\hat{\varphi} = U(\alpha)\hat{\varphi}U(\alpha)^{+} \qquad U(\alpha)|0\rangle = |0\rangle$$

$$\langle 0|U(\alpha)T\hat{\varphi}(x)\hat{\varphi}(y)U^{+}(\alpha)|0\rangle = \langle 0|TU(\alpha)\hat{\varphi}(x)U^{+}(\alpha)U(\alpha)\hat{\varphi}(y)U(\alpha)^{+}|0\rangle$$

$$= \langle 0|Te^{i\alpha}\hat{\varphi}(x)e^{i\alpha}\hat{\varphi}(y)|0\rangle = e^{2i\alpha}\langle 0|T\hat{\varphi}(x)\hat{\varphi}(y)|0\rangle$$

$$iG_F(x-y) = \int \frac{d^4k}{(2\pi\hbar)^4}\cdot\frac{i\hbar c}{(k_\mu/\hbar)^2-(mc/\hbar)^2+i\varepsilon}e^{-ik_\mu(x-y)^\mu/\hbar}$$



時間軸の負の方向に進む負エネルギーの粒子

時間軸の正の方向に進む正エネルギーの反粒子

(Pauli-Weisskopf 半古量子化)

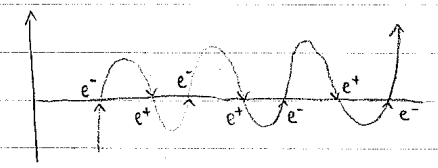
$$\hat{\varphi}(x) = \int \frac{d^3k\,\hbar c}{\sqrt{(2\pi\hbar)^3 2E}}\left[\hat{a}_{\vec{k}}e^{-i(Et-\vec{k}\cdot\vec{x})/\hbar} + \hat{b}^{+}_{\vec{k}}e^{i(Et-\vec{k}\cdot\vec{x})/\hbar}\right]$$

$x>y$

$$\hat{\varphi}^{+}(y)|0\rangle = \int \frac{d^3k\,\hbar c}{\sqrt{(2\pi\hbar)^3 2E}}e^{+i(Et-\vec{k}\cdot\vec{y})/\hbar}\hat{a}^{+}_{\vec{k}}|0\rangle$$

$y>x$

$$\hat{\varphi}(x)|0\rangle = e^{+i(Et-\vec{k}\cdot\vec{y})/\hbar}\hat{b}^{+}_{\vec{k}}|0\rangle$$



Home Work

$\mathcal{L} = \partial_\mu \varphi^*(x)\, \partial^\mu \varphi(x) - \left(\frac{mc}{\hbar}\right)^2 \varphi^*(x)\, \varphi(x)$ から出発して

(i) 正準量子化条件 $\left[\hat{\varphi}(t,\vec{x})^+, \hat{\varphi}(t,\vec{y})\right] = -i\hbar c\, \delta^3(\vec{x}-\vec{y})$ $\left[\hat{\varphi}(t,\vec{x}), \hat{\varphi}(t,\vec{y})\right] = 0$

(ii) $\varphi(x) = \int \frac{d^3k\, \hbar c}{\sqrt{(2\pi\hbar)^3 2E}} \left[\hat{a}_{\vec{k}}\, e^{-i(Et - \vec{k}\cdot\vec{x})/\hbar} + \hat{b}^+_{\vec{k}}\, e^{i(Et-\vec{k}\cdot\vec{x})/\hbar}\right]$

と展開することにより $\left[\hat{a}_{\vec{k}}, \hat{a}^+_{\vec{k}'}\right] = \delta^3(\vec{k}-\vec{k}')$

$\left[\hat{b}_{\vec{k}}, \hat{b}^+_{\vec{k}'}\right] = \delta^3(\vec{k}-\vec{k}')$

$\left[\hat{a}_{\vec{k}}, \hat{b}_{\vec{k}}\right] = 0$ 、 etc.

(iii) エネルギー運動量テンソル

$$T_\nu^\mu(x) = \partial_\nu \varphi^*(x)\, \partial^\mu \varphi(x) + \partial^\mu \varphi^*(x)\, \partial_\mu \varphi(x) - \delta_\nu^\mu \mathcal{L}$$

$$\rightarrow \begin{cases} \partial_\mu T_\nu^\mu = 0 \\ \frac{d}{dt}\int T_\nu^0(x)\, d^3x = 0 \end{cases}$$

(iv) $\varphi(x) \rightarrow e^{i\alpha}\varphi(x)$ に関する Noether Current を導き

$J^\mu(x) = \left(\frac{ie}{\hbar c}\right)\left[\varphi^*(x)\, \partial^\mu \varphi(x) - \partial^\mu \varphi^*(x)\, \varphi(x)\right]$ をみちびき

$Q = \left(\frac{ie}{\hbar c}\right) J^0(x)$ を $\hat{a}_{\vec{k}}$, $\hat{b}_{\vec{k}}$ 等で表わせ

▷ Jun 24

相互作用する場の理論

$\phi(x)$ 実スカラー場 $\phi(x)^* = \phi(x)$ 相互作用

Lagrangian $\mathcal{L} = \frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi - \frac{m_0^2}{2}\phi^2 - \frac{\lambda}{4!}\phi^4$

m_0: 裸の質量 (bare mass

λ: 相互作用定数 (coupling co

以下では $\hbar = c = 1$ (自然単位系)

作用 (action) $S = \int \mathcal{L}\, d^4x$

正準理論

$$\pi_\phi(t,\vec{x}) = \frac{\delta}{\delta\dot{\phi}(t,\vec{x})} S = \partial_0\phi = \dot{\phi}(t,\vec{x})$$

Hamiltonian $\hat{\mathcal{H}} = \pi_\phi \dot{\phi} - \mathcal{L} = \frac{1}{2}\left[\pi_\phi^2 + \sum_{i=1}^{3} \partial_i\phi\,\partial_i\phi + m_0^2\phi^2\right] + \frac{\lambda}{4!}\phi^4 > 0$

量子化 $[\pi_\phi(t,\vec{x}), \phi(t,\vec{y})] = -i\delta(\vec{x}-\vec{y})$ (naive な意味)

$\hat{H} = \int \hat{\mathcal{H}}\, d^3x$; $\frac{d}{dt}\hat{H} = [\hat{H}, \hat{H}] = 0$ $\hat{H}$ は時間に依らない

; $\dot{\hat{\phi}} = [\hat{\phi}, \hat{H}] = i\hat{\pi}_\phi$

; $\dot{\hat{\pi}}_\phi = [\pi_\phi, \hat{H}] = (-i)\left[-\partial_i^2\hat{\phi} + m_0\hat{\phi} + \frac{\lambda}{3!}\phi^3\right]$

$\ddot{\hat{\phi}} - \partial_i\phi + m_0^2\hat{\phi} + \frac{\lambda}{3!}\phi^3 = 0$

$\Box\ddot{\phi}$ ↑ Heisenberg 方程式

D'Alembertian)

$\hat{\phi}(t,\vec{x}) = e^{i\hat{H}t}\,\hat{\phi}(0,\vec{x})\,e^{-i\hat{H}t}$ $(\hbar = 1)$

Heisenberg 表示 Schrödinger 表示

Schrödinger 方程式 $i\frac{\partial}{\partial t}|\psi(t)\rangle = \hat{H}(0)|\psi(t)\rangle$

$\hat{\phi}(0,\vec{x})$ を対角化する表示 ← $\begin{cases} \langle\beta|\hat{\beta} = \beta\langle\beta| \\ \langle\beta|\hat{p} = \frac{1}{i}\frac{\partial}{\partial\beta}\langle\beta| \end{cases}$

$\langle\phi(\vec{x})|\,\hat{\phi}(0,\vec{x}) = \phi(\vec{x})\langle\phi(\vec{x})|$

$\langle\phi(\vec{x})|\,\hat{\pi}_\phi(0,\vec{x}) = \frac{\delta}{\delta\phi(\vec{x})}\langle\phi(\vec{x})|$

$\Psi(t,\phi(\vec{x})) \equiv \langle\phi(\vec{x})|\Psi(t)\rangle \quad\longleftrightarrow\quad \langle\beta|\psi(t)\rangle = \psi(t,\beta)$

$i\frac{\partial}{\partial t}\Psi(t,\phi(\vec{x})) = \langle\phi(\vec{x})|\hat{H}|\Psi(t)\rangle$ Schrödinger 方程式 (functional differential eq)

$= \int d^3y\, \mathcal{H}\left(\phi(\vec{y}), \frac{\delta}{i\delta\phi(\vec{y})}\right)\Psi(t,\phi(\vec{x}))$

$\Psi(t;\phi(\vec{x}_1),\phi(\vec{x}_2),\phi(\vec{x}_3),\cdots)$

一般論にもどって

$i\frac{\partial}{\partial t}|\Psi(t)\rangle = \hat{H}(0)|\Psi(t)\rangle \qquad \hat{H}(0) = \hat{H}_0(0) + \hat{H}_I(0)$

自由部分　相互作用

自由部分

$$\hat{H}_0(0) = \int d^3x \, \frac{1}{2}\left[\hat{\pi}_\phi(\vec{x})\,\hat{\pi}_\phi(\vec{x}) + \sum_{i=1}^{3}\partial_i\phi\,\partial_i\phi + m^2\phi^2\right]$$

$\Downarrow$

繰り込まれた質量　renormalized mass

相互作用部分

$$\hat{H}_I(0) = \frac{\lambda}{4!}\int d^3x\,\hat{\phi}(\vec{x})^4 + \frac{\delta m^2}{2}\int d^3x\,\hat{\phi}(\vec{x})^2$$

↑ mass counter term

$\delta m^2 = m_0^2 - m^2$

$\downarrow \infty \qquad \downarrow \infty$

相互作用表示（Interaction picture）

状態ベクトル $\Psi(t) \to \Psi(t)$ へのユニタリー変換

$$\Psi(t) \equiv e^{i\hat{H}_0 t}\Psi(t) = e^{i\hat{H}_0 t}e^{-i\hat{H}t}\Psi(0)$$

$$i\frac{\partial}{\partial t}\Psi(t) = e^{i\hat{H}_0 t}\left[-\hat{H}_0(0) + \hat{H}(0)\right]e^{-i\hat{H}t}\Psi(0) = \underbrace{e^{i\hat{H}_0 t}\,\hat{H}_I(0)\,e^{-i\hat{H}_0 t}}_{\hat{H}_I(t)}\underbrace{e^{i\hat{H}_0 t}e^{-i\hat{H}t}\Psi(0)}_{\Psi(t)}$$

$$\hat{H}_I(t) = e^{i\hat{H}_0 t}\,\hat{H}_I(0)\,e^{-i\hat{H}_0 t}$$

$$= e^{i\hat{H}_0 t}\int d^3x\,\mathcal{H}_I(\hat{\phi}(0,\vec{x}))\,e^{-i\hat{H}_0 t} = \int d^3x\,\mathcal{H}_I(\phi(t,\vec{x}))$$

すなわち　$\hat{\phi}(t,\vec{x}) = e^{i\hat{H}_0 t}\,\phi(0,\vec{x})\,e^{i\hat{H}_0 t}$

相互作用表示での Schrödinger eq

$$i\frac{\partial}{\partial t}\Psi(t) = \hat{H}_I(t)\Psi(t) \qquad \text{あるいは} \quad \Psi(t) = \Psi(-\infty) - i\int_{-\infty}^{t} dt\,\hat{\mathcal{H}}_I(t)\Psi(t)$$

形式解　$\Psi(t) = T\exp\left[-i\int_{-\infty}^{t}\hat{H}_I(t)dt\right]\Psi(-\infty)$

$$\Psi(+\infty) = T\exp\left[-i\int_{-\infty}^{\infty}\hat{H}_I(t)dt\right]\Psi(-\infty) = S\,\Psi(-\infty) \qquad \begin{pmatrix} c_1(\infty) \\ c_2(\infty) \\ \vdots \end{pmatrix} = S\begin{pmatrix} c_1(-\infty) \\ c_2(-\infty) \\ \vdots \end{pmatrix}$$

S：散乱行列（S-matrix）

$\hat{\phi}(t,\vec{x})$ の従う方程式

$$\hat{\phi}(t,\vec{x}) = e^{i\hat{H}_0 t}\,\hat{\phi}(0,\vec{x})\,e^{-i\hat{H}_0 t}$$

$$\hat{H}_0 \equiv \int d^3x \left[\frac{1}{2}\hat{\pi}_\phi(\vec{x})^2 + \frac{1}{2}\partial_i\hat{\phi}(\vec{x})\,\partial_i\hat{\phi}(\vec{x}) + \frac{1}{2}m^2\hat{\phi}(\vec{x})^2\right]$$

☺ $[\hat{\pi}_\phi(\vec{x}), \hat{\phi}(\vec{y})] = -i\delta^3(\vec{x}-\vec{y})$

$$\dot{\hat{\phi}}(t,\vec{x}) = i e^{i\hat{H}_0 t}\left[\hat{H}_0, \hat{\phi}(0,\vec{x})\right] e^{-i\hat{H}_0 t} = e^{i\hat{H}_0 t}\,\hat{\pi}_\phi(0,\vec{x})\,e^{-i\hat{H}_0 t} = \hat{\pi}_\phi(t,\vec{x})$$

$$\ddot{\hat{\phi}}(t,\vec{x}) = \dot{\hat{\pi}}_\phi(t,\vec{x}) = e^{i\hat{H}_0 t}\, i\left[\hat{H}_0, \hat{\pi}_\phi(0,\vec{x})\right] e^{-i\hat{H}_0 t}$$

$$= e^{i\hat{H}_0 t}\left[\partial_i^2\hat{\phi}(0,\vec{x}) - m^2\hat{\phi}(0,\vec{x})\right] e^{-i\hat{H}_0 t} = \partial_i^2\hat{\phi}(t,\vec{x}) - m^2\hat{\phi}(t,\vec{x})$$

$$\underbrace{(\partial_t^2 - \partial_i^2)}_{\Box}\hat{\phi}(t,\vec{x}) + m^2\hat{\phi}(t,\vec{x}) = 0$$ 場の演算子は自由場の方程式をみたす

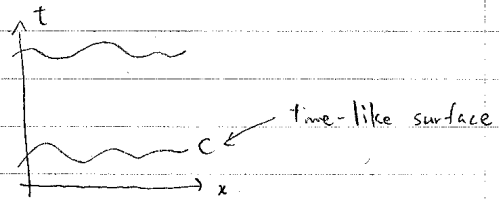
$$\hat{\phi}(t,\vec{x}) = \int \frac{d^3k}{\sqrt{(2\pi)^3 2E}}\left[\hat{a}_{\vec{k}}\, e^{-i(Et-\vec{k}\cdot\vec{x})} + \hat{a}^\dagger_{\vec{k}}\, e^{i(Et-\vec{k}\cdot\vec{x})}\right]$$

但 $E(\vec{k}) = \sqrt{\vec{k}^2 + m^2}$ $\quad \Box e^{\pm i(Et-\vec{k}\cdot\vec{x})} = -(E^2-k^2)e^{\pm i(Et-\vec{k}\cdot\vec{x})}$

$$[\hat{a}_{\vec{k}}, \hat{a}^\dagger_{\vec{k}'}] = \delta^3(\vec{k}-\vec{k}')$$

<u>Lorentz 共変性</u> $\quad i\frac{\partial}{\partial t}\Psi(t) = \hat{H}(t)\Psi(t)$

Tomonaga-Schwinger 式 <u>超多時間理論</u> $\Psi(t_1,\vec{x}_1, t_2,\vec{x}_2, t_3,\vec{x}_3, \cdots)$



量子化

$$[\hat{\phi}(x), \hat{\phi}(y)] = \Delta(x-y)$$

S 行列 $\quad S = T\exp\left[-i\int_{-\infty}^{\infty}\mathcal{H}_I(x)\,d^4x\right]$

$[\hat{H}_I(x), \hat{H}_I(y)] = 0$ for $(x-y)^2 < 0$ 因果律

space-like separation

S行列は特定のLorentz系で計算しなくよい

S行列の要素 $\langle f|S|i\rangle$

状態：Fock空間表示

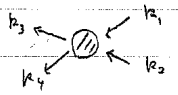
Boson

$$|n_{\vec{k}_1}, n_{\vec{k}_2}, \cdots\rangle = \frac{(\hat{a}^\dagger_{\vec{k}_1})^{n_{\vec{k}_1}}}{\sqrt{n_{\vec{k}_1}!}} \cdots |0\rangle \qquad \vec{k}_1 \text{に} n_{\vec{k}_1}\text{コ}, \vec{k}_2 \text{に} n_{\vec{k}_2}\text{コ}, \cdots$$

S行列の計算 × Feynman 則

$$S = T\exp\left[-i\int_{-\infty}^{\infty}\mathcal{H}_I(x)\,d^4x\right] = 1 - iT\int\mathcal{H}_I(x)\,d^4x + \frac{(-i)}{2!}T\int\mathcal{H}_I(x)\mathcal{H}_I(y)\,d^4x\,d^4y + \cdots$$

例) 2粒子が2粒子へ散乱するS行列の要素



状態ベクトル　始め　$|k_1,k_2\rangle = \hat{a}^\dagger_{\vec{k}_1}\hat{a}^\dagger_{\vec{k}_2}|0\rangle$　　$\hat{a}_k|0\rangle = 0$

終り　$\langle k_3,k_4| = \langle 0|\hat{a}_{\vec{k}_3}\hat{a}_{\vec{k}_4}$　　$\langle 0|\hat{a}^\dagger_k = 0$

$$\langle k_3k_4|S|k_1k_2\rangle = \langle k_3k_4|k_1k_2\rangle + (-i)\langle k_3k_4|T\int\mathcal{H}_I(x)\,d^4x\,|k_1k_2\rangle$$

摂動の0次：$\langle k_3k_4|k_1k_2\rangle = \delta^3(\vec{k}_1-\vec{k}_3)\,\delta^3(\vec{k}_2-\vec{k}_4)$

または　$\delta^3(\vec{k}_1-\vec{k}_4)\,\delta^3(\vec{k}_2-\vec{k}_3)$

素通りする前方散乱

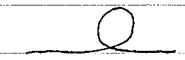
$\mathcal{H}_I = \frac{\lambda}{4!}\phi^4 + \frac{1}{2}\delta m^2\phi^2$　（$\frac{1}{2}\delta m^2\phi^2$ は無視）

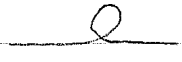
1次：$\langle k_3k_4|\mathcal{H}_I(x)|k_1k_2\rangle = \langle k_3k_4|\frac{\lambda}{4!}\hat{\phi}(x)\hat{\phi}(x)\hat{\phi}(x)\hat{\phi}(x)|k_1k_2\rangle$

$$= \lambda\,\Psi^*_{k_3}(x)\Psi^*_{k_4}(x)\Psi^*_{k_1}(x)\Psi^*_{k_2}(x)$$

$$\Psi_k(x) = \langle 0|\hat{\phi}(x)\hat{a}^\dagger_{\vec{k}}|0\rangle = \frac{1}{\sqrt{(2\pi)^3 2E}}e^{-i(Et-\vec{k}\cdot\vec{x})}$$

▷ Jul 1

$\mathcal{H}_I = \frac{\lambda}{4!}:\hat{\phi}(x)\hat{\phi}(x)\hat{\phi}(x)\hat{\phi}(x):$ ⟶  禁止

$\phi\phi\phi\phi$ ⟶  ゆるされる

$$\langle 0|\phi(x)\, a_k^+|0\rangle = \Psi_k(x) = \frac{1}{\sqrt{(2\pi)^3 2E}} e^{-i(Et-\vec{k}\cdot\vec{x})}$$

$$S = 1 + (-i)\int \mathcal{H}_I\, d^4x + \frac{(-i)^2}{2!} T \int \mathcal{H}_I(x)\mathcal{H}_I(y)\, d^4x d^4y + \cdots$$

$$\langle k_1' k_2' | k_1 k_2\rangle = \delta^3(k_1-k_1')\delta^3(k_2-k_2') \quad \text{or } \delta^3(k_1-k_2')\delta^3(k_2-k_1')$$

$k_1 \longrightarrow k_1'$, $k_2 \longrightarrow k_2'$ / $k_1 \times k_2'$, $k_2 \times k_1'$

摂動の一次の項

$$\hat\phi(x) = \int \frac{d^3k}{\sqrt{(2\pi)^3 2E}} \left[\hat a_{\vec k} e^{-i(Et-\vec k\cdot\vec x)} + \hat a_{\vec k}^+ e^{i(Et-\vec k\cdot\vec x)}\right] \quad \text{に注意すると}$$

$$\langle k_3 k_4|\mathcal{H}_I(x)|k_1 k_2\rangle = \langle k_3 k_4|\frac{\lambda}{4!}:\hat\phi(x)\hat\phi(x)\hat\phi(x)\hat\phi(x):|k_1 k_2\rangle$$

a^+a^+aa

$$= \langle 0|\hat a_{k_3}\hat a_{k_4}\frac{\lambda}{4!}:\hat\phi\hat\phi\hat\phi\hat\phi:\hat a_{k_1}^+\hat a_{k_2}^+|0\rangle$$

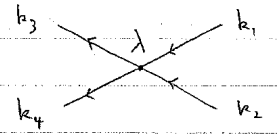
$$= \lambda \Psi_{k_3}^*(x)\Psi_{k_4}^*(x)\Psi_{k_1}(x)\Psi_{k_2}(x)$$

従って $(-i)\langle k_3 k_4|\int d^4x\, \mathcal{H}_I(x)|k_1 k_2\rangle = (-i)\lambda\int d^4x\, \Psi_{k_3(x)}^*\Psi_{k_4}^*(x)\Psi_{k_1}(x)\Psi_{k_2}(x)$

$$= (-i\lambda)\, 2\pi\delta(E_3+E_4-E_1-E_2)\int d^3x\, \Psi_{k_3}^*(\vec x)\Psi_{k_4}^*(\vec x)\Psi_{k_1}(\vec x)\Psi_{k_2}(\vec x)$$

⇒ 量子化による Bose 粒子の扱いで ポテンシャルを $V(\vec x-\vec y) = \lambda\delta^3(\vec x-\vec y)$ と選んだのに対応する

$$\lambda\phi^4(x) \longleftrightarrow \lambda\delta^4(x-y)$$



摂動の二次

$$\frac{(-i)^2}{2!}\langle k_3 k_4|T\int \mathcal{H}_I(x)\mathcal{H}_I(y)\, d^4x\, d^4y\, |k_1 k_2\rangle$$

$$= \frac{(-i)^2}{2!}\int d^4x d^4y\, \langle k_3 k_4|T:\hat\phi(x)^4\hat\phi(y)^4:|k_2 k_3\rangle$$

これを量子論をよく計算するために Wick の定理を使う

Wick の定理

先ず

$$\hat{\phi}(x) = \int \frac{d^3k}{\sqrt{(2\pi)^3 2E}} \left[\hat{a}_{\vec{k}}\, e^{-i(Et - \vec{k}\cdot\vec{x})} + \hat{a}^{\dagger}_{\vec{k}}\, e^{i(Et - \vec{k}\cdot\vec{x})} \right]$$

$\hat{a}_{\vec{k}}\, e^{-i(Et-\vec{k}\cdot\vec{x})} \to \hat{\phi}^{(+)}(x)$: 消滅 + 振動数

$\hat{a}^{\dagger}_{\vec{k}}\, e^{i(Et-\vec{k}\cdot\vec{x})} \to \hat{\phi}^{(-)}(x)$: 生成 − 振動数

このとき (operator / c数)

$$T\,\hat{\phi}(x)\hat{\phi}(y) = :\hat{\phi}(x)\hat{\phi}(y): + \langle 0 | T\,\hat{\phi}(x)\hat{\phi}(y) | 0 \rangle$$

T → 時間順序積

$\langle 0 | T\,\hat{\phi}(x)\hat{\phi}(y)|0\rangle = iG_F(x-y)$

$$T\,\phi(x)\phi(y) = \begin{cases} \phi(x)\phi(y) & x^0 > y^0 \\ \phi(y)\phi(x) & x^0 < y^0 \end{cases}$$

$$:\hat{\phi}(x)\hat{\phi}(x): = \hat{\phi}^{(+)}(x)\hat{\phi}^{(+)}(y) + \phi^+\phi^- + \phi^-\phi^+ + \phi^-\phi^-$$

消滅演算子を常に生成演算子の右へ並べる

例 $x^0 > y^0$ とする

$$T\,\hat{\phi}(x)\hat{\phi}(y) = (\phi^+(x) + \phi^-(x))(\phi^+(y) + \phi^-(y))$$

$$= \phi^+\phi^+ + \phi^-\phi^+ + \phi^+\phi^- + \phi^-\phi^- = :\phi(x)\phi(y): + \hat{\phi}^+(x)\hat{\phi}^-(y) - \hat{\phi}^-(x)\hat{\phi}^+(y)$$

ここで

$$\phi^+\phi^- - \phi^-\phi^+ = \int \frac{d^3k}{\sqrt{(2\pi)^3 2E}} \frac{d^3k'}{\sqrt{(2\pi)^3 2E'}} \left[\hat{a}_{\vec{k}}\hat{a}^{\dagger}_{\vec{k}'} - \hat{a}^{\dagger}_{\vec{k}'}\hat{a}_{\vec{k}} \right] e^{-i(Ex^0 - \vec{k}\cdot\vec{x})}\, e^{i(Ey^0 - \vec{k}'\cdot\vec{y})}$$

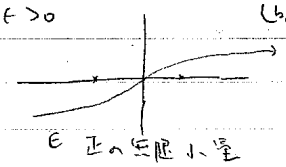
$\delta^3(\vec{k}-\vec{k}')$

$$= \int \frac{d^3k}{(2\pi)^3 2E}\, e^{-iE(x^0+y^0) + i\vec{k}\cdot(\vec{x}-\vec{y})}$$

$k^2 = k_0^2 - \vec{k}^2$

$k_\mu(x^\mu + y^\mu) = k_0(x^0 - y^0) - \vec{k}(\vec{x}-\vec{y})$

$$= \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon}\, e^{-ik_\mu(x^\mu - y^\mu)}$$



$$= \langle 0 | T\,\hat{\phi}(x)\hat{\phi}(y) | 0 \rangle = iG_F(x-y)$$

$T\hat\phi(x_1)\hat\phi(x_2)\cdots\hat\phi(x_n) = :\phi(x_1)\phi(x_2)\cdots\phi(x_n): + \langle 0|T\hat\phi(x_1)\hat\phi(x_2)|0\rangle :\hat\phi(x_3)\cdots\hat\phi(x_n): + \text{Permutation}$

$+\langle 0|T\hat\phi(x_1)\phi(x_3)|0\rangle\langle 0|T\hat\phi(x_2)\hat\phi(x_4)|0\rangle :\hat\phi(x_5)\cdots\hat\phi(x_n):$

$+\langle 0|T\hat\phi(x_1)\phi(x_2)|0\rangle\cdots\langle 0|T\hat\phi(x_{n-1})\phi(x_n)|0\rangle \quad (n:\text{even})$

$+\langle 0|T\phi(x_1)\phi(x_2)|0\rangle\cdots\langle 0|T\phi(x_{n-2})\phi(x_{n-1})|0\rangle\hat\phi(x_n) \quad (n:\text{odd})$

Wick の定理を証明する為には x_{n+1} を一番右に選んで

$:\hat\phi(x_1)\cdots\hat\phi(x_n):\hat\phi(x_{n+1}) = :\hat\phi(x_1)\hat\phi(x_2)\cdots\hat\phi(x_n)\hat\phi(x_{n+1}): + \sum_j :\phi(x_1)\cdots\check{\phi}(x_j)\cdots\phi(x_n): \langle 0|T\hat\phi(x_j)\hat\phi(x_{n+1})|0\rangle$

となる $\hat\phi(x_{n+1})$ については自明に成り立っていることを予め

[証明]

A と B を $\{1,2,\cdots,n\}$ の互いに異なる二つの組の和 $A\cup B$ への分割とする

$$:\hat\phi(x_1)\cdots\hat\phi(x_n): = \sum_A \prod_{i\in A}\phi^{(-)}(x_i)\prod_{j\in B}\phi^{(+)}(x_j)$$

従って $:\hat\phi(x_1)\cdots\hat\phi(x_n)\hat\phi(x_{n+1}): = :\hat\phi(x_1)\cdots\hat\phi(x_n):(\phi^{(+)}(x_{n+1})+\phi^{(-)}(x_{n+1}))$

$= :\phi(x_1)\cdots\phi(x_n)\phi^{(+)}(x_{n+1}): + :\phi(x_1)\cdots\phi(x_n):\phi^{(-)}(x_{n+1})$

次に $:\hat\phi(x_1)\cdots\hat\phi(x_n):\hat\phi^{(-)}(x_{n+1}) = \sum_A\prod_{i\in A}\phi^{(-)}(x_i)\prod_{j\in B}\phi^{(+)}(x_j)\hat\phi^{(-)}(x_{n+1})$

$= \sum_A\prod_{i\in A}\phi^{(-)}(x_i)\phi^{(-)}(x_{n+1})\prod_{j\in B}\phi^{(+)}(x_j) + \sum_A\prod_{i\in A}\phi^{(-)}(x_i)\sum_{j'\in B}\phi^{(+)}(x_{j'})\langle 0|T\hat\phi(x_{j'})\hat\phi(x_{n+1})|0\rangle$

$= :\phi(x_1)\cdots\phi(x_n)\phi^{(-)}(x_{n+1}): + \sum_{j'}:\phi(x_1)\cdots\check{\phi}(x_{j'})\cdots\phi(x_n):\langle 0|T\hat\phi(x_{j'})\hat\phi(x_{n+1})|0\rangle$

従って帰納的に ↓ これで Wick の定理の証明 終わり 7/15

$:\phi(x_1)\cdots\phi(x_n):\phi(x_{n+1}) = :\phi(x_1)\cdots\phi(x_n)\phi(x_{n+1}): + \sum_j :\phi(x_1)\cdots\check{\phi}(x_j)\cdots\phi(x_n):\langle 0|T\hat\phi(x_j)\hat\phi(x_{n+1})|0\rangle$

例

$T\phi(x_1)\phi(x_2) = :\phi(x_1)\phi(x_2): + \langle 0|T\phi(x_1)\phi(x_2)|0\rangle$

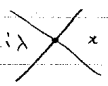
$(T\phi(x_1)\phi(x_2))\phi(x_3) = :\phi(x_1)\phi(x_2)\phi(x_3): + \langle 0|T\phi(x_1)\phi(x_2)|0\rangle\phi(x_3)$ $\quad x_3^0 < x_1^0, x_2^0$

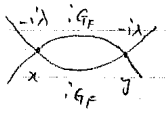
$T\phi(x_1)\phi(x_2)\phi(x_3) = :\phi(x_1)\phi(x_2)\phi(x_3): + \phi(x_1)\langle 0|T\phi(x_2)\phi(x_3)|0\rangle +$

$+\phi(x_2)\langle 0|T\phi(x_1)\phi(x_3)|0\rangle +$

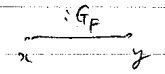
S行列を計算するFeynman則

頂点 $\mathcal{H}_I(x) = \frac{\lambda}{4!}\hat{\phi}(x)^4$





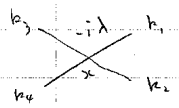
伝搬関数 $iG_F(x-y) = \langle 0|T\hat{\phi}(x)\hat{\phi}(y)|0\rangle$



(i) 位相的(topological)に区別できるダイアグラムを全て正しく書き上げる
頂点の位置に関しては積分する

(ii) ダイアグラムの外線が始状態あるいは終状態になる
$\Psi_k(x)$ 始
$\Psi_k^*(x)$ 終 } を用いる

(iii) 適当な対称因子で割る



$$\int d^4x\,(-i\lambda)\Psi^*_{k_3}(x)\Psi^*_{k_4}(x)\Psi_{k_1}(x)\Psi_{k_2}(x)$$

▷ Jul 8

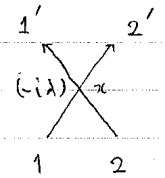
摂動計算

$$S = T\exp\left[-i\int_{-\infty}^{\infty} H_{int}(x)\,d^4x\right]$$

$\langle f|S|i\rangle$ = $|i\rangle$, $\langle f|$: Fock state

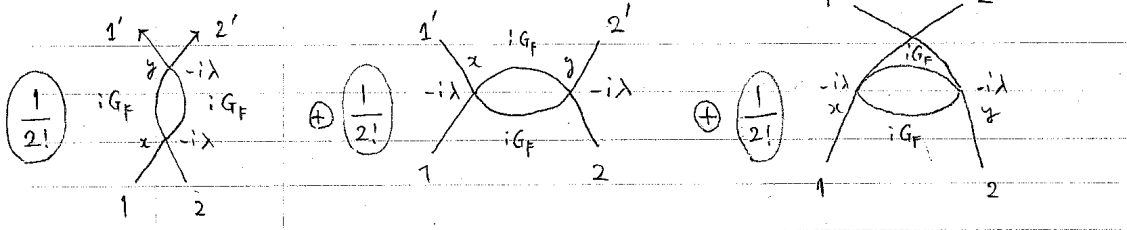
Feynman図

(1) $1+2 \to 1'+2'$
最低次(λに関して)



$$\Rightarrow \text{振幅} = (-i\lambda)\int_{-\infty}^{\infty} d^4x\,\Psi^*_{k_1'}(x)\Psi^*_{k_2'}(x)\Psi_{k_1}(x)\Psi_{k_2}(x)$$

(2) 2次の項 (λ^2)



$\Psi = \langle 0|\hat{\phi}\, a^+|0\rangle$

$$\Rightarrow \left(\frac{1}{2!}\right)(-i\lambda)^2 \int d^4x\, d^4y \Big[\Psi^*_{k_1'}(y)\Psi^*_{k_2'}(y)\, iG_F(x-y)\, iG_F(x-y)\, \Psi_{k_1}(x)\Psi_{k_2}(x) \quad \leftarrow \text{未1図}$$

$$+ \Psi^*_{k_1'}(x)\Psi^*_{k_2'}(y)\, iG_F(x-y)\, iG_F(x-y)\, \Psi_{k_1}(x)\Psi_{k_2}(y) \quad \leftarrow \text{未2図}$$

$$+ \Psi^*_{k_1'}(y)\Psi^*_{k_1'}(x)\, iG_F(x-y)\, iG_F(x-y)\, \Psi_{k_1}(x)\Psi_{k_2}(y) \Big] \quad \leftarrow \text{未3図}$$

(1) 頂点 $(-i\lambda)\int x$

(2) 内線 $x \cdot\!\!-\!\!-\!\!-

$\overbrace{\phi(x)\phi(y)} = \langle 0|T\phi(x)\phi(y)|0\rangle$

$(4C_2)^2$ = 4コから2コ取り出す組合せの積

$2!$ = 2コ(x座標) × 2コ(y座標)の演算子から $\langle 0|T\phi(x)\phi(y)|0\rangle$ をつくる組合せ

$T\phi^4(x)\phi^4(y)$

$= \cdots + (4C_2)(4C_2)2!\overbrace{\phi(x)\phi(x)\overbrace{\phi(y)\phi(y)}} : \phi(x)\phi(x)\phi(y)\phi(y) :$

$\int d^4x d^4y$

$S = 1 + \cdots + \frac{(-i)^2}{2!} TH_I(x)H_I(y) + \cdots$

$\frac{(-i\lambda)^2}{2!}\frac{1}{(4!)^2} T\phi^4(x)\phi^4(y)$ $\quad\Big\Big T\phi^4(x)\phi^4(y)\,d^4x\,d^4y$

over all 4y factor　相互作用項

Dyson展開　$\frac{1}{2!}\left(\frac{-i\lambda}{4!}\right)^2 \times :\phi^4(x)\phi^4(y):$

レポート

(i) 3つの項の導出

(ii) 3つの項がそれぞれ何ケセットを与えるか

(iii) 対応するFeynman図とFeynman則

$\frac{(-i\lambda)^2}{2!}\frac{1}{(4!)^2}(4C_2)^2 \times 2!$

$= \frac{1}{2!}\left(\frac{-i\lambda}{2!}\right)^2 \times \frac{1}{2!}$

$\frac{(-i\lambda)^2}{2!}\frac{1}{(4!)^2}(4C_3)^2$

$= \frac{1}{2!}\left(\frac{-i\lambda}{3!}\right)^2$

$\begin{cases} \frac{1}{2!}\left(\frac{-i\lambda}{3!}\right)^2 \times \langle 0|T\phi(x)\phi(y)|0\rangle :\phi^3\phi^3: \\ \frac{1}{2!}\left(\frac{-i\lambda}{2!}\right)^2\left(\frac{1}{2!}\right) \times \langle\ \rangle\langle\ \rangle :\phi^2\phi^2: \\ \frac{1}{2!}\left(\frac{-i\lambda}{2!}\right)^2\left(\frac{1}{3!}\right) \times \langle\ \rangle\langle\ \rangle\langle\ \rangle :\phi\phi: \end{cases}$



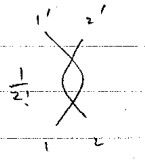
$\frac{1}{2!}(-i\lambda)^2\left(\frac{1}{4!}\right) \times \langle\ \rangle\langle\ \rangle\langle\ \rangle\langle\ \rangle$

次にFeynman則と比べるには $\frac{(-i\lambda)^2}{2!}\left(\frac{1}{2!}\right)\langle 0|T\phi(x)\phi(y)|0\rangle\langle 0|T\phi(x)\phi(y)|0\rangle \times \underline{\langle k_1' k_2'|:\phi^2(x)\phi^2(y):|k_1 k_2\rangle}$

$\langle 0|a_{k_1'}a_{k_2'}:\phi^2(x)\phi^2(y):a^\dagger_{k_1}a^\dagger_{k_2}|0\rangle$

$= \left(\frac{-i\lambda}{2!}\right)^2\frac{1}{2!}(2!)^2\langle 0|T\phi(x)\phi(y)|0\rangle\langle 0|T\phi(x)\phi(y)|0\rangle$

$\times \Psi^*_{k_1'}(x)\Psi^*_{k_2'}(x)\Psi_{k_1}(y)\Psi_{k_2}(y)$



$= (-i\lambda)^2\underbrace{\frac{1}{2!}}_{\text{重複度}}\langle 0|T\phi(x)\phi(y)|0\rangle\langle 0|T\phi(x)\phi(y)|0\rangle \cdot \Psi^*_{k_1'}(x)\Psi^*_{k_2'}(x)\Psi_{k_1}(y)\Psi_{k_2}(y)$

$\underbrace{iG_F(x-y)}$

くりこみ

$$\int d^4\xi_1 d^4\xi_2 \int d^4x\, d^4y \; \frac{e^{i\xi_1(x-y)}}{\xi_1^2 - m^2 + i\varepsilon} \cdot \frac{e^{i\xi_2(x-y)}}{\xi_2^2 - m^2 + i\varepsilon} \; e^{ik_1' x + ik_2' x - ik_1 y - ik_2 y}$$

$$= \int d^4\xi_1 d^4\xi_2 \;\; \delta^4(\xi_1 - \xi_2 \quad\quad) \, \delta^4(\quad\quad) \; \frac{1}{\xi_1^2 - m^2} \frac{1}{\xi_2^2 - m^2}$$

$$= \int d^4\xi_1 \int d^4\xi_2$$

$$= \lambda^2 \, \delta^4(k_1' + k_2' - k_1 - k_2) \int d^4\xi_1 \; \frac{1}{\xi_1^2 - m^2} \cdot \frac{1}{(\xi_1 + k_1' + k_2')^2 - m^2}$$ ultra-violet log 発散

↓ 発散する項 $\sim \lambda^2 \, \delta^4(k_1' + k_2' - k_1 - k_2) \log \Lambda^2$

これと一次の項

$$\lambda \, \delta^4(k_1' + k_2' - k_1 - k_2) \qquad (\lambda + \lambda^2 \ln \Lambda^2) \, \delta^4(k_1' + k_2' - k_1 - k_2)$$

くりこまれて $\lambda_r = \lambda + \lambda^2 \ln \Lambda^2$

Dirac 方程式

	スピン	
scalar field	0	実 / 複素
spinor field	1/2	Mejorana 場 / Dirac 場
vector field	1	A_μ Maxwell
tensor field	2	$g_{\mu\nu}$ Einstein

Spinor

$\psi_\alpha(x)$ $\alpha = 1, 2, 3, 4$ spinor の足

$$\psi(x) = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}$$ 4成分の場

Pauli 行列の一般化 γ-matrix

γ^μ $\mu = 1, 2, 3, 4$ 時空間の足

4行4列の行列

$\gamma^0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ $\gamma^k = \begin{pmatrix} 0 & \tau^k \\ -\tau^k & 0 \end{pmatrix}$ $k = 1,2,3$

τ^k : Pauli 行列 $\tau^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ $\tau^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ $\tau^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$(\gamma^0)^\dagger = \gamma^0$ $(\gamma^k)^\dagger = -\gamma^k$

γ^μ の一般的な定義 $\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu} \rightarrow \begin{cases} (\gamma^0)^2 = 1 \\ (\gamma^1)^2 = -1 \\ \gamma^0\gamma^1 + \gamma^1\gamma^0 = 0 \end{cases}$

$g^{\mu\nu} = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}$ Minkowski

Dirac 方程式

$(i\gamma^\mu\partial_\mu - m)\psi(x) = 0$

$\mathcal{L} = \bar{\psi}(x)[i\gamma^\mu\partial_\mu - m]\psi(x)$

$\bar{\psi} = \psi^\dagger\gamma^0$

$\bar{\psi}(x)$ に関する変分 $\delta\bar{\psi}$ → Dirac-eq

この式に左から $(i\gamma^\mu\partial_\mu + m)$ を掛けると $(\gamma^\mu\partial_\mu)(\gamma^\nu\partial_\nu) = -\gamma^\mu\gamma^\nu\partial_\mu\partial_\nu$

$= -\frac{1}{2}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu)\partial_\mu\partial_\nu$

$= -g^{\mu\nu}\partial_\mu\partial_\nu = -\partial_\mu\partial^\mu = -\Box$

$(i\gamma^\mu\partial_\mu + m)(i\gamma^\mu\partial_\mu - m)\psi = 0$

$(i\gamma^\mu\partial_\mu)^2 - m^2$ $(\Box + m^2)\psi = 0$

Klein-Gordon

"Dirac 方程式は Klein-Gordon 方程式の square-root をとったものと見なせる"

Lorentz 不変性

$\psi'_\alpha(x') = S_{\alpha\beta}\psi_\beta(x)$ $(x')^\mu = a^\mu{}_\nu x^\nu$

(Report) (学部) (i) $1+2 \to 1'+2'$ 散乱の振幅を Feynman 則を用いて導き
打消因子の origin を説明せよ
(ii) $T\phi(x)\phi(y) = \langle 0|T\phi(x)\phi(y)|0\rangle + :\phi(x)\phi(y):$ を示せ

Dirac equation

Lorentz 不変性 $S = \int \bar{\psi}(x)(i\gamma^\mu \partial_\mu - m)\psi(x)\, d^4x$ $\quad \bar{\psi}(x) = \psi^+(x)\gamma^0$

$\psi_\alpha(x)$ $\alpha = 1, 2, 3, 4$

Lorentz 変換 $x'^\mu = a^\mu{}_\nu x^\nu$

$x'^\mu x'_\mu = \text{inv.} = g_{\mu\nu} x'^\mu x'^\nu$

Spinor の変換則

$$\psi'_\alpha(x') = \sum_\beta S_{\alpha\beta}\psi_\beta(x)$$

作用の変化は

$$\int \bar{\psi}'(x')(i\gamma^\mu \partial'_\mu - m)\psi'(x')\, d^4x'$$

$$= \int \bar{\psi}(x)\gamma^0 S^+ \gamma^0 [i\gamma^\mu \partial_\mu - m] S\psi(x)\, d^4x$$

$\to \gamma^0 S^+ \gamma^0 = S^{-1}$ ← 質量項の不変性

$S^{-1}\gamma^\mu S = a^\mu{}_\nu \gamma^\nu$

$\to S^{-1}\gamma^\mu \partial'_\mu S = a^\mu{}_\nu \gamma^\nu \partial'_\mu = \gamma^\mu \partial_\mu$

$\left(x'^\mu = a^\mu{}_\nu x^\nu \to \dfrac{\partial}{\partial x'^\mu} \right.$
$\left. x^\mu \partial_\mu = x'^\mu \partial'_\mu \right)$

$\because)\ x^\mu \partial_\mu (x)^2 = 2x^\mu x_\mu = 2(x)^2$

このような S が見つかれば $\int \bar{\psi}'(x')[i\gamma^\mu \partial'_\mu - m]\psi'(x')\, d^4x$

$= \int \bar{\psi}(x)(i\gamma^\mu \partial_\mu - m)\psi(x)\, d^4x$ ⟩ 作用の Lorentz 不変性

S の具体形

$x'^\mu = a^\mu{}_\nu x^\nu = [g^\mu{}_\nu - \underline{\Delta\omega^\mu{}_\nu}]x^\nu$

無限小

$\Delta\omega_{\mu\nu} = g_{\mu\alpha}\Delta\omega^{\alpha}{}_{\nu} = -\Delta\omega_{\nu\mu}$ であり (回転の一般化)

$\Rightarrow S = \exp\left[\frac{1}{2}\Delta\omega_{\mu\nu}S^{\mu\nu}\right]$, $S^{\mu\nu} = \frac{1}{4}[\gamma^{\mu},\gamma^{\nu}]$

∵ γ^{μ} の定義式 $\{\gamma^{\mu},\gamma^{\nu}\}_{+} \equiv \gamma^{\mu}\gamma^{\nu}+\gamma^{\nu}\gamma^{\mu} = 2g^{\mu\nu}$ より

$$\left[\underbrace{\tfrac{1}{4}[\gamma^{\mu},\gamma^{\nu}]}_{S^{\mu\nu}},\gamma^{\alpha}\right] = -i\left(g^{\mu\alpha}\gamma^{\nu}-g^{\nu\alpha}\gamma^{\mu}\right)$$

$$S^{\dagger} = \exp\left[-\tfrac{1}{2}\Delta\omega_{\mu\nu}\underbrace{(S^{\mu\nu})^{\dagger}}_{\downarrow S^{\mu\nu}}\right]$$

$\left(\frac{1}{4}[\gamma^{\mu},\gamma^{\nu}]\right)^{\dagger}$

$= -\frac{1}{4}\left[(\gamma^{0}\gamma^{1})^{\dagger}-(\gamma^{1}\gamma^{0})^{\dagger}\right] = -\frac{1}{4}\left[(\gamma^{1})^{\dagger}(\gamma^{0})^{\dagger}-(\gamma^{0})^{\dagger}(\gamma^{1})^{\dagger}\right]$

$= -\frac{1}{4}\left[-\gamma^{1}\gamma^{0}+\gamma^{0}\gamma^{1}\right] = -\frac{1}{4}\left[\gamma^{0}\gamma^{1}-\gamma^{1}\gamma^{0}\right]$

$= -\frac{1}{4}[\gamma^{1},\gamma^{0}] = -\frac{1}{4}\left[\gamma^{0}\gamma^{1}-\gamma^{1}\gamma^{0}\right]$

$= \frac{1}{4}[\gamma^{1},\gamma^{0}]$

$\gamma^{0}(S^{\mu\nu})^{\dagger}\gamma^{0} = S^{\mu\nu}$

$\downarrow$

$\gamma^{0}S^{\dagger}\gamma^{0} = S^{-1}$

$S^{\dagger}\gamma^{\alpha}S = \left(1-\frac{1}{2}\Delta\omega_{\mu\nu}S^{\mu\nu}\right)\gamma^{\alpha}\left(1+\frac{1}{2}\Delta\omega_{\mu\nu}S^{\mu\nu}\right) = \gamma^{\alpha}-\frac{1}{2}\Delta\omega_{\mu\nu}[S^{\mu\nu},\gamma^{\alpha}]$

$= \gamma^{\alpha}-\frac{1}{2}\Delta\omega_{\mu\nu}\left(g^{\mu\nu}\gamma^{\nu}-g^{\nu\alpha}\gamma^{\mu}\right) = \gamma^{\alpha}-\Delta\omega^{\alpha}{}_{\nu}\gamma^{\nu} = (\gamma^{\alpha})'$

特に z 軸の回りのこの回転は $\Delta\omega_{12} = \varepsilon = -\Delta\omega_{21}$

回転の生成子 (generator)

$S^{3} = S^{12} = \frac{1}{4}[\gamma^{1},\gamma^{2}] = \frac{1}{2}\begin{bmatrix}\tau^{3} & 0\\ 0 & \tau^{3}\end{bmatrix}$

$\because$ 4次元 Pauli 理論 ($\sigma^{3} = \frac{1}{2}\tau^{3}$, etc) の一般化になる

Dirac の作用にもどって $\int \bar{\psi}(x)(i\gamma^\mu \partial_\mu - m)\psi(x)\, d^4x$

$$= \int \psi^\dagger(x)\gamma^0\left[i\gamma^\mu\partial_\mu - m\right]\psi(x)\, d^4x$$

$$= \int \psi^\dagger(x)\left[i\frac{\partial}{\partial t} + i\gamma^0\gamma^k\frac{\partial}{\partial x^k} - \gamma^0 m\right]\psi(x)\, d^4x$$

$$= \int \psi^\dagger(x)\left[i\partial_t - h\right]\psi\, d^4x$$

第一量子化の Hamiltonian $h = -i\gamma^0\gamma^k\partial_k + m\gamma^0$ 但 $\begin{cases}\alpha^k = \gamma^0\gamma^k & k=1,2,3\\ \beta = \gamma^0\\ p^k = \frac{1}{i}\frac{\partial}{\partial x^k}\end{cases}$

$$= \frac{1}{i}\gamma^0\gamma^k\partial_k + m\gamma^0$$

$$= \alpha^k p^k + m\beta$$

静止している場合は

$$h = \beta m = \begin{pmatrix} m & & & \\ & m & & \\ & & -m & \\ & & & -m \end{pmatrix}$$

この正のエネルギー解と2つの負のエネルギー解の4つの独立解がある

$$u_1 = \begin{pmatrix}1\\0\\0\\0\end{pmatrix} \quad u_2 = \begin{pmatrix}0\\1\\0\\0\end{pmatrix} \quad v_1 = \begin{pmatrix}0\\0\\1\\0\end{pmatrix} \quad v_2 = \begin{pmatrix}0\\0\\0\\1\end{pmatrix}$$

$$\begin{cases} hu = mu \\ hv = -mv \end{cases}$$

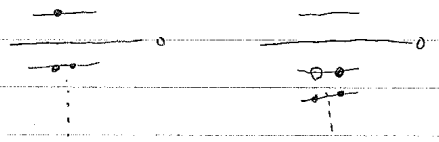
Dirac の hole 理論

物理的真空 ｛負のエネルギー状態が全て occupy されている｝

素粒に



一粒子状態　　一反粒子状態



真空状態を基準にして考えると

エネルギー	$-E$	粒子の欠損	エネルギー	E
運動量	$-\vec{p}$		運動量	$\vec{p}$
スピン	$\downarrow$		スピン	$\uparrow$

反粒子の存在

例えば



$\gamma \longrightarrow e^+e^-$

Dirac 方程式の正のエネルギー解

$$\psi(x) = e^{-ip_\mu x^\mu} u(p) = e^{-i(Et-\vec{p}\cdot\vec{x})} u(p)$$

$(p_0 = E > 0)$

$(i\not{\partial} - m)\psi(x) = 0$ に代入 $\quad \not{\partial} \equiv \gamma^\mu \partial_\mu$

↓

$$(\not{p} - m)u(p) = 0$$

$$\not{p} = E\gamma^0 - \vec{p}\vec{\gamma} = \begin{pmatrix} E & -\vec{p}\cdot\vec{\sigma} \\ \vec{p}\vec{\sigma} & -E \end{pmatrix}$$

すなわち

$u = \begin{pmatrix} \xi \\ \eta \end{pmatrix}$ と2成分 (spinor) ξ, η を用いると

$$\begin{pmatrix} E-m & -\vec{p}\cdot\vec{\sigma} \\ \vec{p}\vec{\sigma} & -E-m \end{pmatrix}\begin{pmatrix} \xi \\ \eta \end{pmatrix} = 0 \Rightarrow \begin{cases} E = \sqrt{|\vec{p}|^2 + m^2} \\ \eta = \dfrac{\vec{p}\vec{\sigma}}{E+m}\xi \end{cases} \text{と同等である}$$

$$u(p, S^z = \tfrac{1}{2}) = \sqrt{E+m}\begin{pmatrix} \xi \\ \dfrac{\vec{p}\vec{\sigma}}{E+m}\xi \end{pmatrix} \qquad \xi = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ up-state}$$

$-\frac{1}{2}$ $\qquad\qquad$ $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$

負のエネルギー解

$(k_0 = E > 0)$

$$\psi(x) = e^{ik_\mu x^\mu} v(k) = e^{i(Et - \vec{k}\cdot\vec{x})} v(k)$$

$$(i\gamma^\mu \partial_\mu \psi(x) = 0 \quad (\not{k} + m) v(k) = 0 \quad \begin{bmatrix} E+m & k\tau \\ k\tau & -E+m \end{bmatrix} \begin{bmatrix} \eta \\ \zeta \end{bmatrix} = 0 \quad \Rightarrow \quad \begin{cases} E = \sqrt{k^2 + m^2} \\ \zeta = \dfrac{k\tau}{E+m}\eta \end{cases}$$

$$v(k, s_z = \tfrac{1}{2}) = \sqrt{E+m} \begin{pmatrix} \frac{k\tau}{E+m}\eta \\ \eta \end{pmatrix} \qquad \eta = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \qquad\qquad = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$-\tfrac{1}{2}$$

直交性と規格条件

$$\bar{u}(k,s) = u(k,s)^\dagger \gamma^0$$

$$\bar{v}(k,s) = v(k,s)^\dagger \gamma^0$$

$$\begin{cases} \bar{u}(k,s)\, u(k,s') = 2m\, \delta_{ss'} \\ \bar{v}(k,s)\, v(k,s') = -2m\, \delta_{ss'} \\ \bar{u}(k,s)\, v(k,s') = 0 \end{cases} \qquad \gamma = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$$

$$\begin{cases} \sum_s u_\alpha(k,s)\, \bar{u}_\beta(k,s) = (\not{k} + m)_{\alpha\beta} \\ \sum_s v_\alpha(k,s)\, \bar{v}_\beta(k,s) = (\not{k} - m)_{\alpha\beta} \end{cases}$$

$$\begin{cases} u^\dagger(k,s)\, u(k,s') = \bar{u}(k,s)\gamma^0 u(k,s') = 2E\, \delta_{ss'} \\ v^\dagger(k,s)\, v(k,s') = \bar{v}(k,s)\gamma^0 v(k,s') = 2m\, \delta_{ss'} \\ u^\dagger(k,s)\, v(k_0, -\vec{k}, s') = \bar{u}(k,s)\gamma^0 v(k_0, -\vec{k}, s') = 0 \end{cases}$$

$$\begin{cases} (\not{k} - m) u(k) = 0 \\ \bar{u}(k)(\not{k} - m) = 0 \end{cases}$$

$$\bar{u}\gamma^0 \not{k} u = m \bar{u}\gamma^0 u$$

$$\bar{u}\not{k}\gamma^0 u = m \bar{u}\gamma^0 u$$

$$\bar{u}\{\gamma^0, \not{k}\} u = 2m \bar{u}\gamma^0 u$$

$$2E\, \bar{u} u = 2m\, \bar{u}\alpha^0 u$$

完全系条件

$$\sum_s \left\{ u_\alpha(k,s)\, u^\dagger_\beta(k,s) + v_\alpha(k_0, -k, s)\, v^\dagger_\beta(k_0, -k, s) \right\} = 2E\, \delta_{\alpha\beta}$$

場の量子化

$$\pi_\psi(x) = \frac{\delta}{\delta \dot{\psi}(x)} \int \psi(x)^\dagger \gamma^0 (i\gamma^\mu \partial_\mu - m) \psi(x)\, d^4x = i\psi^\dagger(x)$$

量子化条件 (fermi 粒子)

$$\{\hat{\psi}_\alpha(t,\vec{x}),\ \hat{\pi}_\beta(t,\vec{y})\} = i\delta^3(\vec{x}-\vec{y})\delta_{\alpha\beta}$$

$$\downarrow$$

$$\{\hat{\psi}_\alpha^\dagger(t,\vec{x}),\ \hat{\psi}_\beta(t,\vec{y})\}_+ = \delta^3(\vec{x}-\vec{y})\delta_{\alpha\beta}$$

$$\{\psi^\dagger,\ \psi^\dagger\} = 0$$

$$\hat{\psi}_\alpha(x) = \int \frac{d^3k}{\sqrt{(2\pi)^3 2E}} \sum_s \left[\hat{b}(\vec{k},s)\, u_\alpha(k,s)\, e^{-ik_\mu x^\mu} + \hat{d}^\dagger(\vec{k},s)\, v_\alpha(k,s)\, e^{ik_\mu x^\mu} \right]$$

$(k_0 = E > 0)$

$$\Rightarrow \{\hat{b}(\vec{k},s),\ \hat{b}^\dagger(\vec{k}',s')\}_+ = \{\hat{d}(\vec{k},s),\ \hat{d}^\dagger(\vec{k}',s')\}_+ = \delta_{ss'}\delta^3(\vec{k}-\vec{k}')$$

他の組み合わせ e.g. $\{\hat{b}^\dagger, \hat{b}^\dagger\} = 0$

Hamiltonian $\mathcal{H} = \pi_\psi(x)\dot{\psi}(x) - \mathcal{L} = \psi^\dagger(x)(-i\gamma^0\gamma^k\partial_k + \gamma^0 m)\psi(x)$

$$\hat{H} \equiv \int d^3x\, \hat{\psi}^\dagger(x)(-i\gamma^0\gamma^m\partial_m + \gamma^0 m)\hat{\psi}(x)$$

$$= \int d^3k \sum_s \left\{ E(k)\, \hat{b}^\dagger(\vec{k},s)\hat{b}(\vec{k},s) - E(k)\, \hat{d}(\vec{k},s)\hat{d}^\dagger(\vec{k},s) \right\}$$

$$= \int d^3k \sum_s \left\{ E(k) \left\{ \hat{b}^\dagger(\vec{k},s)\hat{b}(\vec{k},s) + \hat{d}^\dagger(\vec{k},s)\hat{d}(\vec{k},s) \right\} \right.$$

$$\left. - \sum_s \int d^3k\, E(k) \right\}$$

$\sim\sim\sim\sim$ ← 全てのエネルギー状態の和　負の

$$\Rightarrow :H: = \sum_s \int d^3k\, E(k) \left(b^\dagger b + d^\dagger d \right)$$

この表式は Dirac の hole 理論の考え方と一致する

粒子描像

電子 $|e^-(k,s)\rangle = \hat{b}^\dagger(k,s)|0\rangle$　　$b|k,s\rangle|0\rangle = 0$

陽電子 $|e^+(k,s)\rangle = \hat{d}^\dagger(k,s)|0\rangle$　　$d(k,s)|0\rangle = 0$

$(b^\dagger)^2|0\rangle = 0$ fermi 統計

$\hat{H} |e^-(k,s)\rangle = E(k) |e^-(k,s)\rangle$

$\hat{H} |e^+(k,s)\rangle = \epsilon(k) |e^+(k,s)\rangle$